

FSD - Fee Application Module (FAM)

Functional Specification Document (FSD)

Fee Application Module (FAM)

Document Details

Versi on No.	Change Summary	Reviewed By & Date	Approved By & Date
0.1	Initial draft	Marston Business Team Mar 9, 2026	
0.2	Latest BRD changes (business and technical)	Marston Business Team	
0.3	Added current client setting details for CMF reference		

Table of Contents

Functional Specification Document (FSD)

Fee Application Module (FAM)

Table of Contents

1. Executive Summary

1.1 Purpose

1.2 Scope

1.3 Key Objectives

1.4 Document Structure

1.2 Scope

1.3 Key Objectives

1.4 Document Structure

2. Scope of the Fee Application Module (FAM)

2.1 In Scope

2.2 Out of Scope

2.3 Supported Debt Types

2.4 Current State Analysis

2.4.1 Existing Columbus Fee Implementation

2.4.2 Current Limitations

1. Hard-coded values

2. No Single Source of Truth (SSOT)

3. No effective-date versioning

4. No uplift or graduated fee logic

5. No client-specific overrides

6. No dual-control governance

7. Limited audit

8. No compliance period management

9. No support for fee bubbles

10. Limited support for optimise

11. No structured integration with downstream reporting

Summary

- 3. Functional Requirements
 - 3.1 Fee Table Management (FR-01)
 - Description
 - Functional Requirements
 - 3.2 Fee Uplift Calculation (FR-02)
 - Description
 - Functional Requirements
 - 3.3 Interest Rate Configuration (FR-03)
 - Description
 - Functional Requirements
 - 3.4 Compliance Period Management (FR-04)
 - Description
 - Functional Requirements
 - 3.5 CMF Integration (FR-05)
 - Description
 - Client Case Type (CCT) Settings - Complete Reference Guide3.5.1 Overview
 - Quick Reference Summary
 - 3.5.2 General Settings
 - Basic Identification
 - Contact Information
 - Batch Configuration
 - 3.5.3 Operations Settings
 - Workflow & Processing
 - Case Timing & Expiration
 - Hold Configuration
 - Case Assignment & Matching
 - VRM & Vehicle Processing
 - Referral & Team Permissions
 - Return & Reinstatement
 - Goods & Assets
 - DVLA & Address Verification
 - Additional Operation Settings
 - 3.5.4 Finance Settings
 - Fee & Cost Limits
 - Payment Configuration
 - Payment Distribution Options (PaymentDistributionDto Enum)
 - Linked Cases Money Distribution Options (LinkedCasesMoneyDistribution Enum)
 - Arrangement Settings
 - Direct Payment Settings
 - Interest Settings
 - Successful Completion Fee
 - 3.5.5 Remittance & Invoicing Settings
 - Currency Configuration
 - Payment Run Frequencies
 - Payment Flags
 - 3.5.6 Client Web (CWA) Integration Settings
 - Client Web Access (CWA) Settings
 - 3.5.6 Tracing Settings
 - Address Change Notifications
 - COA (Change of Address) Configuration
 - Enforce at New Address
 - COA Letter Templates
 - COA Verifier Conditions
 - COA Verifier Configuration (CoAExistingVerifiersDTO)
 - COA Process Chain
 - Process Flow Overview
 - COA Verification Purposes
 - Standard COA Verifier (CoaVerifier)
 - Custom COA Verifier (CustomCoaVerifier)
 - COA Fail Reasons (DefaulterActionFailReason Enum)
 - COA Actions Performer
 - COA Configuration Setting
 - Trace Processing
 - Expired Cases
 - 3.5.8 Configuration Mappings
 - Configuration Mapping Entity
 - Configuration Categories
 - 3.5.9 Validation Rules
 - Required Fields
 - Length Constraints
 - 9.3 Numeric Constraints
 - Conditional Requirements
 - 3.5.10. Related Entities

Tab wise details	Functional Requirements
3.6	Keyer-Verifier Workflow (FR-06)
	Description
	Functional Requirements
3.7	Fee Bubble Logic (FR-07)
	Description
	Functional Requirements
3.8	Optimise Parity & Conflict Handling (FR-08)
	Description
	Functional Requirements
3.9	VAT Handling (FR-09)
3.10	Versioned Activation (FR-10)
5.	Architecture Design
5.1	High-Level Architecture
	A. Configuration Layer (SSOT)
	B. Processing Layer
	C. Output Layer
4.2	Logical Architecture Diagram
4.3	Component Responsibilities
4.3.1	Configuration Layer Components
	Fee Table
	Fee Uplift Table
	Interest Table
	Compliance Period Table
	CMF Integration
	Keyer-Verifier Workflow
	Audit Store
	Fee Table Management (FR-01)
	Fee Attributes
	Supported Fee Stages
	Example Fee Table Structure
4.4	Processing Layer Components
4.4.1	Fee Calculation Engine
4.4.2	Uplift Engine
	Uplift Calculation Formula
4.4.3	Interest Engine
	Interest Table Structure
	Tier Selection Logic
4.4.4	Compliance Period Engine
	Compliance Period Table
	Greater-Of Logic
4.4.5	Pro-Rata Allocation Engine
4.4.6	Fee Bubble Selector
4.4.7	Rule Combination Engine
4.4.8	Fee Versioning & Effective Date Handling
	Versioning Model
	Version States
	Version Timeline
	Supersession Logic
	Future-Dated Configuration
	Maximum Future Date Range
4.5	Output Layer Components
4.5.1	Fee APIs
4.5.2	Audit Outputs
4.5.3	Integration Connectors
4.5.4	Case & Balance Updates
4.6	Architecture Principles
6.	Data Models
5.1	FeeTableEntity
5.2	FeeUpliftEntity
5.3	InterestRateEntity
5.4	CompliancePeriodEntity
5.5	CMFOVERRIDEEntity (Logical Model)
5.6	ChangeRequestEntity
5.7	AuditTrailEntity
5.8	FeeCalculationResult (Logical Model)
6.	API Specifications
6.1	Fee Calculation API
	Request:
	Response:
6.2	Fee Configuration API
7.	Flow Diagrams
7.1	Fee Calculation Flow

- 7.2 Keyer-Verifier Approval Flow
- 7.3 Interest Calculation Flow
- 7.4 Client Override Logic
- 7.5 Debt Aggregation Process
- Sequence Flows for Key Operations
- 7.6 Fee Calculation Sequence
- 7.7 Configuration Approval Sequence
- 8. Security & Access Control
- 8.1 Security Principles
 - 1. Least Privilege Access
 - 2. Segregation of Duties
 - 3. Immutable Audit
 - 4. Controlled Configuration Lifecycle
 - 5. Full Traceability
- 8.2 Roles & Responsibilities
 - 1. Viewer
 - 2. Keyer (Proposer)
 - 3. Verifier
 - 4. Administrator
- 8.3 Permission Matrix
- 8.4 Segregation of Duties
- 8.5 Approval Workflow Security Controls
 - 8.5.1 State Machine Enforcement
 - 8.5.2 Immutability
 - 8.5.3 Automatic Activation
- 8.6 Authentication & Authorization
 - 8.6.1 Authentication
 - 8.6.2 Authorization
- 8.7 Audit Security
- 8.8 Notification Security
- 8.9 Optimise & Device Security
- 8.10 Security Constraints
- 9. Validation Rules
 - 9.1 Fee Table Validation Rules
 - 9.2 Uplift Validation Rules
 - 9.3 Interest Rate Validation Rules
 - 9.4 Compliance Period Validation Rules
- 10. Business Rules
 - 10.1 Fee Application Rules
 - 10.2 Aggregation Rules
 - 10.3 Greater-Of Rules
 - 10.4 Fee Bubble Rules
- 11. Audit Requirements
 - 11.1 Audit Principles
 - 1. Immutability
 - 2. End-to-End Traceability
 - 3. Source Transparency
 - 4. Non-Repudiation
 - 5. Automatic Capture
 - 11.2 Audit Record Structure
 - 11.3 Audit Events
 - A. Configuration Lifecycle
 - B. Fee Calculation Events
 - C. CMF Override Events
 - D. Workflow Actions
 - E. Exception & Correction Events
 - 11.4 Audit Notification Requirements
 - Notifications Triggered At
 - Notification Recipients
 - 11.5 Audit Storage Requirements
 - 11.6 Audit for Fee Bubble Logic
 - 11.7 Audit for Compliance Period Logic
 - 11.8 Audit for Interest Calculation
 - 11.9 Audit for Uplift Calculation
- 12. Integration Requirements
 - 12.1 Integration Principles
 - 1. Server-Side Authoritative Logic
 - 2. Deterministic Responses
 - 3. Version-Aware Processing
 - 4. SSOT Referencing Only
 - 5. Audit Coverage
 - 6. Clear Error & Exception Handling
 - 7. Backwards Compatible API Contracts

- 12.2 optimise Integration
 - 12.2.1 Online Path
 - 12.2.2 Offline Path
 - 12.2.3 Configuration Sync
 - 12.2.4 Direct-Apply Flag
- 12.3 CMF Integration
 - 12.3.1 Required CMF Data
 - 12.3.2 Precedence Rule
 - 12.3.3 Audit Traceability
- 12.4 KeyIVR Integration
 - 12.4.1 Payment Flow Integration
 - 12.4.2 Arrangement Validations
 - 12.4.3 Real-Time Validation
- 12.5 Reporting & Data Warehouse
 - 12.5.1 Required Outputs
 - 12.5.2 ETL Requirements
 - 12.5.3 Reconciliation Support
- 12.6 BizTalk / Messaging Integrations
 - 12.6.1 Required Fields in Payloads
 - 12.6.2 Backwards Compatibility
- 12.7 Event Grid Integration
 - 12.7.1 Event Types
 - 12.7.2 Requirements
- 12.8 External Client Portals / CWA
 - Requirements
- 12.9 Reconciliation & Finance Integrations
 - Requirements
- 12.10 Error Handling & Failover Behavior
 - Error Types
 - Failover Requirements
- 12.11 Integration Non-Functional Requirements
 - Performance
 - Scalability
 - Monitoring
- 13. Implementation Priority
 - 13.1 Phase 1 — CORE (High Priority)
 - Scope of Phase 1
 - Outcome
 - 13.2 Phase 2 — Enhancements (Medium Priority)
 - Scope of Phase 2
 - Outcome
 - 13.3 Phase 3 — Integration & Extensions (High Priority)
 - Scope of Phase 3
 - Outcome
 - 13.4 Phase 4 — Advanced Capabilities (Optional / Priority-Based)
 - Scope of Phase 4
 - Outcome
 - 13.5 Implementation Principles
- 14. Data Retention & Destruction
 - 14.1 Retention Policy Overview
 - 6 years from the date of case closure
 - 14.2 Data Categories Covered
 - 1. Fee Configuration History
 - 2. Workflow Records
 - 3. Audit Trail Entries
 - 4. Fee Calculation Snapshots
 - 5. CMF Override Retrieval Records
 - 6. Integration Records
 - 14.3 Purpose of Retention
 - A. Regulatory & Statutory Compliance
 - B. Financial Accuracy
 - C. Audit Assurance
 - D. Complaints & Disputes Handling
 - 14.4 Data Storage Requirements
 - Secure, Immutable Storage
 - Version Preservation
 - Indexing & Performance Considerations
 - Disaster Recovery
 - Access Controls
 - 14.5 Destruction Policy
 - Secure & Irreversible Deletion
 - Records Eligible for Destruction
 - Destruction Safeguards
 - Destruction Logging

- 14.6 Compliance Considerations
- 15. Risks & Mitigations
 - 15.1 Risk Register
 - Risk 1 — Activation mismatch across estate or optimise devices
 - Risk 2 — In-flight case ambiguity
 - Risk 3 — Incorrect debt aggregation for uplift
 - Risk 4 — Distribution & commission ripple effects
 - Risk 5 — Fee Bubble version misalignment
 - Risk 6 — Missing or incorrect CMF overrides
 - Risk 7 — Workflow bottlenecks (Keyer → Verifier)
 - Risk 8 — Audit volume & performance
 - Risk 9 — Reporting inconsistencies
 - Risk 10 — optimise offline mismatch
 - Risk 11 — User error in manual transitions
 - Risk 12 — Pro-rata allocation misapplication
 - Risk 13 — Incorrect interest tier selection
 - Risk 14 — Failure to enforce immutability
 - Risk 15 — Future-dated activation errors
 - Risk 16 — Operational misunderstandings of new rules
 - Risk 17 — Legal/Regulatory ambiguities
 - 15.2 Summary of Risk Treatment Strategy

1. Executive Summary

1.1 Purpose

The Fee Application Module (FAM) introduces a configuration-driven, fully governable, and auditable framework for managing all enforcement fees across all supported debt streams. It replaces legacy hard-coded logic with a modern, flexible, centrally-controlled architecture that ensures deterministic, transparent, and compliant fee calculations for every case.

FAM consolidates all fee-related parameters — including base fees, uplift thresholds and percentages, interest tiers, compliance periods, CMF overrides, fee versioning, and audit controls — into a Single Source of Truth (SSOT). All configuration changes follow a dual-control approval process (Keyer → Verifier 1 → Verifier 2) to ensure governance, segregation of duties, and complete traceability. Each change is versioned, future-date supported, and automatically activated at the specified effective window without code deployment.

The module provides deterministic fee outcomes by applying consistent rules for fee selection, uplift logic, debt aggregation, pro-rata distribution, VAT handling, and client-specific overrides. It also ensures correct behavior under special scenarios such as fee bubbles, future-dated fee versions, cross-linked cases, High Court interest, optimise device parity, and offline-sync reconciliation.

End-to-end enforcement journey behavior is fully supported. FAM evaluates case stage, eligibility conditions, compliance period status, client configuration, and audit requirements before applying any fee. It ensures that Compliance Fee is applied at NOE issuance, uplift rules use aggregated debt within the same debt group, compliance periods honor the greater-of rule (statutory vs CMF), and interest tiers respect client overrides.

The solution also delivers full lifecycle auditability, including immutable before/after values, digital signatures, user role tracking, change rationale, approval chain history, and automatic notifications to the Audit function. All fee calculations, overrides, CMF retrievals, and bubble selections generate audit artefacts suitable for operational, financial, and regulatory review.

FAM integrates seamlessly with ecosystem components including Columbus, optimise (online/offline), CMF, KeyIVR, reporting layers, and downstream teams. It exposes APIs for fee calculation, configuration management, audit retrieval, and compliance period evaluation. Reporting and Data Warehouse pipelines consume the FAM SSOT and fee model indicators to ensure accurate financial outputs and historical traceability.

Overall, the Fee Application Module establishes a controlled, flexible, auditable, and future-proof foundation for all enforcement fee management activities, reducing operational risk, improving compliance, and enabling rapid policy adoption across all enforcement streams.

1.2 Scope

The FAM replaces legacy hard-coded fee handling with a configuration-first, audit-controlled platform. All fee policy elements including base fees, uplifts, interest rates, and compliance periods are maintained in a Single Source of Truth (SSOT) governed by dual-control approvals.

1.3 Key Objectives

Objective	Description
-----------	-------------

Accuracy	Statutory fees applied correctly and consistently across all enforcement debt types
Governance	RBAC-controlled configuration tables with dual-person approval (Keyer-Verifier)
Compliance	Full audit trails for all configuration changes with automatic notifications
Flexibility	Business users can configure fees without IT intervention
Integration	Seamless reference to Client Maintenance File (CMF) for client-specific parameters
Transparency	All fee calculations are deterministic, explainable, and auditable

1.4 Document Structure

This document is organized into distinct sections covering:

- Functional requirements and business rules
- Technical architecture and component design
- Data models and configuration tables
- Integration points and API specifications
- Security, audit, and compliance controls
- Sequence flows and worked examples

1.2 Scope

The FAM replaces legacy hard-coded fee handling with a configuration-first, audit-controlled platform. All fee policy elements including base fees, uplifts, interest rates, and compliance periods are maintained in a Single Source of Truth (SSOT) governed by dual-control approvals.

1.3 Key Objectives

Objective	Description
Accuracy	Statutory fees applied correctly and consistently across all enforcement debt types
Governance	RBAC-controlled configuration tables with dual-person approval (Keyer-Verifier)
Compliance	Full audit trails for all configuration changes with automatic notifications
Flexibility	Business users can configure fees without IT intervention
Integration	Seamless reference to Client Maintenance File (CMF) for client-specific parameters
Transparency	All fee calculations are deterministic, explainable, and auditable

1.4 Document Structure

This document is organized into distinct sections covering:

- Functional requirements and business rules
- Technical architecture and component design
- Data models and configuration tables
- Integration points and API specifications

- Security, audit, and compliance controls
- Sequence flows and worked examples

2. Scope of the Fee Application Module (FAM)

2.1 In Scope

The following capabilities are within scope for the FAM implementation:

Category	Components
Configuration Management	SSOT for fees, uplifts, interest, compliance periods
Approval Workflow	Dual-control Keyer → Verifier1 → Verifier2 workflow
Fee Calculation	Deterministic base fee, uplift, VAT, interest calculation
Compliance	Compliance period greater-of logic (statutory vs client)
Extensions	Compliance extensions with configurable limit checks
Advanced Logic	Pro-rata allocation, aggregation logic, fee bubble versioning
Integration	Mobile/Optimise parity, offline mismatch detection, server override
Audit	Immutable audit trail, notifications, 6-year retention
Reporting	Fee model/version indicators for Data Warehouse

2.2 Out of Scope

The following are explicitly excluded from FAM scope:

Category	Exclusion
Core Workflow	Changes to workflow engine beyond fee-related transitions
New Debt Types	Introduction of new enforcement stages or debt types
Payment Logic	Redesign of debt recovery or payment arrangement logic
Commission	Commission model redesign outside fee-value impacts
Legacy Systems	Legacy system decommissioning
UI Features	New Optimise UI features beyond fee application
External Systems	New external integrations

2.3 Supported Debt Types

The FAM supports the following enforcement debt types:

Debt Stream	Case Type ID	Case Type Name	Group
Magistrates & Traffic Enforcement	8	Magistrates Distress (inc. Inland Revenue)	WOC
Magistrates & Traffic Enforcement	9	RTA/TMA (Road Traffic / Traffic Management)	WOC
Arrest & Committal Enforcement	1	Arrest Bail	Arrest
Arrest & Committal Enforcement	2	Arrest No Bail	Arrest
Arrest & Committal Enforcement	3	Arrest Breach Bail	Arrest
Arrest & Committal Enforcement	17	Arrest Committal	Arrest

Arrest & Committal Enforcement	18	Arrest Section 136	Arrest
Arrest & Committal Enforcement	19	Arrest Breach No Bail	Arrest
Local Authority & Civil Debt	20	Council Tax Liability Order	Liability Order
Local Authority & Civil Debt	21	National Non Domestic Rates	Liability Order
Local Authority & Civil Debt	22	Child Maintenance Group	Liability Order
Local Authority & Civil Debt	23	Commercial Rent	Liability Order
Local Authority & Civil Debt	24	Legal Aid	Liability Order
Local Authority & Civil Debt	6	Debt Collection	Liability Order
Local Authority & Civil Debt	11	Clamping Orders	Liability Order
High Court Enforcement	25	High Court	High Court

2.4 Current State Analysis

2.4.1 Existing Columbus Fee Implementation

The current Columbus enforcement fee implementation relies heavily on legacy constructs that were designed for simpler, static statutory fee models. While these components have supported operations historically, they are no longer sufficient for a modern, configurable, governed fee management framework.

Key system components include:

- **SchemeCharge**

Stores fee and charge configurations but lacks income-stream classification, versioning, effective-date logic, tiering structures, and governance controls.

- **ChargeApplication**

Records fee applications to cases but does not capture version identifiers, rule sources, effective windows, CMF overrides, or fee-bubble decisions.

- **EnforcementFee**

Holds fixed enforcement fee values but is not scalable for multi-version, multi-stage, or multi-stream fee structures.

- **InterestRate / ResumeInterestRate**

Interest is partially supported but relies on hardcoded logic, lacks SSOT tables, and does not support client-specific override rules or tiered thresholds.

- **ChargeFacade / Calculation Logic**

Fee calculation logic is embedded in code, limiting agility and preventing business-driven configuration changes. It also lacks uplift rules, greater-of logic, compliance-period evaluation, fee-bubble selection, and audit traceability.

Overall, Columbus currently uses a **code-driven, non-governed fee model**, with limited visibility and inflexible configuration capabilities.

2.4.2 Current Limitations

The existing fee architecture presents several critical limitations that the FAM module resolves:

1. Hard-coded values

- Fee amounts, thresholds, percentages, interest rates, and compliance periods are embedded in code.
- Any regulatory or policy change requires development effort, testing cycles, and deployments.
- No ability to manage future-dated configurations.

2. No Single Source of Truth (SSOT)

- Fee rules, uplift percentages, interest tiers, and compliance periods are scattered across logic, constants, and database tables.
- No unified fee repository for audit, reporting, or governance.

3. No effective-date versioning

- The system cannot maintain multiple versions of fee tables.
- No mechanism to determine which version applies for a given case event date.
- Historical fee accuracy cannot be guaranteed.

4. No uplift or graduated fee logic

- Current design cannot handle uplift thresholds, graduated percentages, or aggregated debt-based calculations.
- No support for debt aggregation across linked cases.

5. No client-specific overrides

- CMF integration does not exist.
- Cannot support client-specific:
 - Compliance periods
 - Fee caps
 - Interest overrides
 - Custom parameters

6. No dual-control governance

- No Keyer → Verifier1 → Verifier2 approval workflow.
- No segregation of duties or governance to protect fee configuration integrity.

7. Limited audit

- No immutable audit records for configuration changes.
- No before/after value tracking or digital signatures.
- No audit notifications to governance teams.

8. No compliance period management

- Compliance period is fixed at 14 days.
- No greater-of logic (statutory vs CMF).
- No automatic expiry and transition enforcement.
- No differentiated UI presentation.

9. No support for fee bubbles

- Systems cannot determine oldest fee version across linked cases.
- Inadequate handling of mixed-version groups and mid-bubble version changes.

10. Limited support for optimise

- Device parity is not guaranteed.
- Offline scenarios cannot detect and reconcile mismatches.
- No server-side override logic for fee discrepancies.

11. No structured integration with downstream reporting

- Reporting and Data Warehouse cannot distinguish fee versions or fee models.
- No Fee Model Indicator for TCE vs FAM.

Summary

The current Columbus fee infrastructure does not satisfy modern governance, transparency, auditability, explainability, or configuration-driven needs.

FAM addresses all these limitations by introducing:

- A governed SSOT framework
- Version-controlled fee configuration
- Deterministic, rule-driven fee calculation
- Audit-secure change management

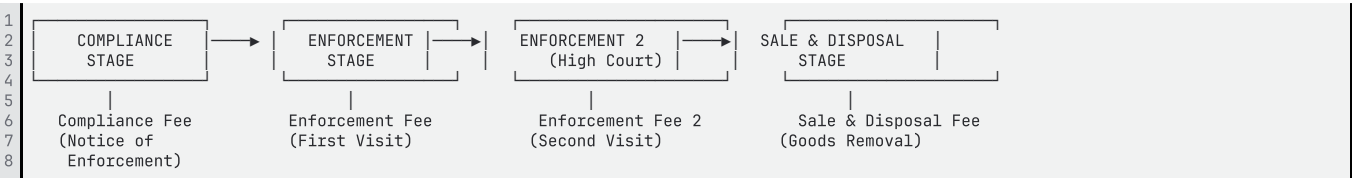
- CMF-aware client-specific behavior
- End-to-end compliance and uplift logic
- optimise online/offline parity
- Reporting and Data Warehouse alignment

This establishes the foundation for accurate, scalable, flexible, and fully audited fee management across all enforcement streams.

3. Functional Requirements

This section defines the functional behaviors, business rules, configuration structures, approval workflows, and calculation logic required for the Fee Application Module (FAM).

The enforcement journey follows a structured progression through distinct stages, each with specific fee implications:



3.1 Fee Table Management (FR-01)

Description

The system shall maintain a fully configurable, version-controlled Fee Table as the Single Source of Truth (SSOT) for all enforcement fee stages across all debt types. Fee Table entries must support future-dated activation, historical preservation, dual-control approvals, and deterministic fee retrieval based on case event date.

Functional Requirements

FR-01.01 — Fee Attributes

The system shall store fees with the following attributes:

- Income Stream
- Fee Stage (Compliance, Enforcement, Enforcement 2 where applicable, Sale & Disposal)
- Fee Amount
- Effective Start Date
- Effective End Date (optional; open-ended if latest version)
- Status (Draft, Pending Approval, Active, Retired)
- Created By / Created Date
- Verified By (Verifier 1 and Verifier 2) / Approved Dates
- Rationale for change
- Configuration version identifier

FR-01.02 — Effective Date Versioning

The system shall support multiple versions of fee configurations and shall automatically select the correct version based on:

- Case event date
 - Fee stage
 - Income stream
- Only one active version per stage per income stream shall apply at any given time.

FR-01.03 — Future-Dated Configuration

Future fee versions may be created and approved in advance.
They shall automatically activate at 00:00 on their effective date without manual intervention.

FR-01.04 — No Overlaps

The system shall ensure that effective date ranges do not overlap within the same Income Stream + Fee Stage combination.

FR-01.05 — Supported Income Streams

The system shall support all enforcement debt types defined in business policy (e.g., Local Authority, RTA/TMA, Magistrates, High Court, Commercial Debt, etc.).

FR-01.06 — Supported Fee Stages

The system shall support all required statutory fee stages:

- Compliance
- Enforcement
- Enforcement 2 (High Court)
- Sale & Disposal

FR-01.07 — Immutable History

Historical fee versions must remain immutable, auditable, and available for calculation replay.

3.2 Fee Uplift Calculation (FR-02)

Description

The system shall calculate uplift fees for eligible fee stages using uplift thresholds, uplift percentages, rounding rules, and aggregated debt logic maintained in the SSOT.

Functional Requirements

FR-02.01 — Uplift Table Structure

The uplift table shall include:

- Income Stream
- Fee Stage
- Uplift Threshold
- Uplift Percentage
- Rounding Rules
- Effective Start/End Dates
- Approval metadata

FR-02.02 — Threshold Validation

Uplift applies only when the aggregated debt exceeds the configured threshold for the relevant stage.

FR-02.03 — Aggregated Debt Logic

The system shall aggregate debts across all *active linked cases* belonging to the same debtor and same debt group. Closed cases and cases from other streams must be excluded.

FR-02.04 — Uplift Formula

$$\text{Uplift} = (\text{Aggregated Debt} - \text{Threshold}) \times \text{Uplift \%}$$

Rounding must apply according to configuration (e.g., always round up).

FR-02.05 — Multiple Fee Versions Support

Uplift configurations follow the same versioning rules as the Fee Table.

FR-02.06 — Enforcement and S&D Support

Uplift rules shall apply to Enforcement fees, Sale & Disposal fees, or any other fee stages flagged as uplift-eligible.

3.3 Interest Rate Configuration (FR-03)

Description

The system shall support tiered interest calculations with both standard interest table entries and client-specific override rates retrieved from CMF.

Functional Requirements

FR-03.01 — Tiered Interest Table

The interest table shall define:

- Balance thresholds
- Tiered interest rates
- Effective date ranges
- Approval metadata

FR-03.02 — Highest Applicable Tier Rule

For each case, the system shall apply the rate corresponding to the highest threshold the balance meets.

FR-03.03 — Client Override Precedence

If CMF provides a client-specific interest override, that value takes full precedence over the standard table.

FR-03.04 — Future Version Activation

New interest configurations shall activate automatically based on effective date.

FR-03.05 — Audit

Each interest evaluation shall record whether a table value or client override was applied.

3.4 Compliance Period Management (FR-04)

Description

The system shall calculate and enforce compliance periods using statutory base values and client-specific CMF values, applying a greater-of selection rule.

Functional Requirements

FR-04.01 — Compliance Period Table

The SSOT must store compliance periods per income stream with effective dates and approval metadata.

FR-04.02 — CMF Integration (Greater-Of Rule)

Compliance period = the greater of:

- Statutory compliance period
- CMF-configured client period

FR-04.03 — Phase-Based UI Behavior

Case details must show:

- Data Cleanse Phase view
- Compliance Phase (Statutory) view
- Compliance Phase (CMF) view
- Comparison View (both periods)

FR-04.04 — Expiry Trigger

On compliance period expiry, the system automatically transitions the case to Enforcement stage and applies the Enforcement Fee.

FR-04.05 — Audit Flags

Audit must capture:

- Statutory value
- CMF value
- Value selected
- Source used

3.5 CMF Integration (FR-05)

Description

The system shall retrieve client-specific parameters from CMF and apply appropriate overrides for fee caps, interest, compliance periods, and additional client policies.

Client Case Type (CCT) Settings - Complete Reference Guide

3.5.1 Overview

The **Client Case Type (CCT)** is a core configuration entity in the Columbus system that defines how cases are handled for a specific client and case type combination. Each CCT represents a unique combination of:

- A **Client** (the enforcement agency or company)
- A **Case Type - Debt Stream** (e.g., High Court, County Court, Traffic)

The CCT settings control every aspect of case handling including:

- Case lifecycle and expiration
- Payment arrangements and fee structures
- Remittance and invoicing processes
- Address change handling
- Client Web (CWA) integrations
- Workflow templates and automation

Quick Reference Summary

Tab	Key Settings
General	Case Type, Name, Brand, Office, Batch Reference Prefix, Contact Info
Operations	Workflow, Charging Scheme, Life Expectancy, Client Life Expectancy, Auto Assign, Auto Link, VRM, DVLA
Finance	Fee Cap, Cost Cap, Min Fees, Payment Scheme, Payment Distribution, Arrangements
Remittance	Currencies, Payment Frequencies, Client/Officer Paid flags, Invoice Run
Tracing	Address Change notifications, COA settings, Enforce at New Address, Trace via Workflow
Client Web (CWA)	Show Notes, Show History, Show Attachments in Final Only

3.5.2 General Settings

The General tab contains basic identification and contact information for the Client Case Type.

Client Settings

Transport for London CC

General

Operations

Payments and Charges

Remittance and Invoicing

Client Web

Tracing

Case Type

RTA/TMA (RoadTraffic/Traffic Management)

Abbreviated Name

TfL CC New - 1147

Brand

Marston TMA & HMCTS

Office

Bridgman House

Batch Confirmation Email

Enforcementagents@tflenforcement.com

Batch Reference Prefix

TfL CC New

Primary Contact Name

Wayne Gainsburgh

Primary Contact Address

230 Blackfriars Road, London SE1 8NW

Notification Email Address

Useful Phone Numbers

Contact Centre

03333202988

Automated Line

03333201100

Copy Settings

Save

Close

Basic Identification

Property	Type	Length	Required	Description
CaseType	CaseType	•	Yes	Determines which Case Type this Client Case belongs to. This is the primary classification of cases (e.g., HighCourt, CountyCourt, Traffic).
Name	string	1-80	Yes	Abbreviated Name - A short name/alias for the Client Case Type. While filled in for each client, it is not used across the company.
Brand	Brand	•	Yes	Determines which Brand the Client's cases belong to. Used for branding and organizational purposes.
Office	Office	•	Yes	Location - The physical office or location responsible for handling these cases.
BatchRefPrefix	string	1-50	Yes	Batch Reference Prefix - This setting determines a prefix for each Batch of Cases created for the given client. Used in batch identification and reporting.

Contact Information

Property	Type	Length	Description
ContactName	string	1-256	Contact Name - Primary contact person for this Client Case Type.
ContactAddress	string	1-500	Contact Address - Physical address for the client contact.
CrossReferenceEmail	string	max 512	Cross Reference Email - Email address used for cross-referencing. Used for communication and integration with client systems. Can contain multiple emails separated by semicolons.

NotificationEmailAddress	string	•	Notification Email - Email address where notifications about case activities are sent.
ContactCentrePhoneNumber	string	max 50	Contact Centre Phone Number - Phone number for the contact centre handling these cases.
AutomatedLinePhoneNumber	string	max 50	Automated Line Phone Number - Phone number for automated services.
StandardIndustryCode	string	max 50	Standard Industry Code - Industry classification code for the client.

Batch Configuration

Property	Type	Description
Batch Confirmation Email	string	Under this address, a message is sent which informs the client that batches of Cases have been loaded onto the source system. This is an automated notification to confirm batch processing completion.

3.5.3 Operations Settings

The Operations tab controls case handling, workflow, timing, and permissions.

Client Settings

Transport for London CC

General

Operations

Payments and Charges

Remittance and Invoicing

Client Web

Tracing

Client Workflow

Workflow

TMA_TFL

Charging Scheme

RTA/TMA Standard Charging Scheme

Apply Charge on Assignment

Default Hold Time Period

365 days

Case Age

Life Expectancy

365 days

Client Life Expectancy

1 days

Extend due to Arrangement

1 days

Extend due to Trace

1 days

Expired Case Return Code

Standard Industry Code

Expired cases Assignable

Reinstate

Permit Goods Removed

Allow Direct Hold with Life ext. past Expiry

Auto Return Expired Cases

Use Decisioning via Workflow

Officer remain assigned for Hold cases

Officer remain assigned for EB or Goodwill refund type on cases

2nd Referral Work

Auto Assign Matches

Use Locksmith card

Enable Automatic Linking

Enable Manual Linking

Enable Nulla Bona

Include Unattended Returns

Returned Case Number Cap

0

Permit VRM

Permit Welfare

Permit Financial Difficulty Referral

Permit Return Case Prematurely

Permit Return for a Part-Paid case

Permit DVLA Enquiries

Auto recheck after

84 days

Notify New DVLA Address

Use Message Exchange

Exclude Marston Payments

Allow return of cases with supplementary indicator status

Outbound Letter Calendar

Letter Actions Wait on Holiday Dates

Copy Settings

Save

Close

Workflow & Processing

Property	Type	Default	Description
Workflow	string	•	Workflow Template - This setting determines which Workflow Template is associated with the given client. Defines the stages and actions a case goes through.

ChargingScheme	ChargingScheme	Required	Charging Scheme - This setting determines which scheme of Charges is associated with the given Client. Defines how fees are calculated and applied.
UseMessageExchange	bool	false	Use Message Exchange - Enables integration with the message exchange system for case communications.
ResponseFile	bool	false	Response File - Determines whether response files should be generated for this client.
BlockLinkedGantCasesProcessingViaWorkflow	bool	false	Block Linked GANT Cases Processing Via Workflow - When enabled, prevents linked GANT cases from being processed through workflow automation.
IsDecisioningViaWorkflowUsed	bool	false	Use Decisioning Via Workflow - Enables automated decision-making through workflow processes.

Case Timing & Expiration

Property	Type	Default	Valid Range	Description
LifeExpectancy	int	Required	1-9999	Life Expectancy - This setting determines the maximum age of a case. After this number of days, the case becomes expired in the system. This is the standard case lifecycle duration.
ClientLifeExpectancy	int?	null	1-9999	Client Life Expectancy - When enabled, this setting determines how long a Client expects us to work on a Warrant before taking further action. This overrides the standard Life Expectancy for client-specific workflows.
DaysToExtendCaseDueToArrangement	int?	null	1-9999	Extend due to Arrangement - When "Client Life Expectancy" is enabled, this setting indicates how many "extension" days a Client should be informed about for extensions based on Active Payment Arrangement.
DaysToExtendCaseDueToTrace	int?	null	1-9999	Extend due to Trace - When "Client Life Expectancy" is enabled, this indicates how many "extension" days a Client should be informed about for extensions based on positive outcome of trace activities.
DefaultHoldTimePeriod	int	0	1-999	Default Hold Time Period - This setting determines a default number of days a Client's Case will be held when hold is performed without explicitly specifying a period.
ReturnCap	int?	null	0-10000	Return Cap - A limit on the number of cases that can be returned for this client type. Used to control return volume.
CurrentReturnCap	int?	null	0-10000	Current Return Cap - The current value of the return cap tracking.

Hold Configuration

Property	Type	Default	Description
IsDirectHoldWithLifeExtPastExpiryAllowed	bool	false	Allow Direct Hold with Life Extension Past Expiry - This setting determines whether the given Client can be held past the Expiry Date. Enables holding cases beyond their normal expiration.
IsNullBonaRequired	bool	false	Nulla Bona Required - Specifies whether Nulla Bona (no goods found) status is required for case completion.

Case Assignment & Matching

Property	Type	Default	Description
AssignMatches (Auto Assign Matches)	bool	false	Auto Assign Matches - This setting determines whether Client's Cases should be assigned to an officer together with their assignable, not assigned linked Cases by AutoAssignMatchesProcessor. When enabled, linked cases are automatically assigned together.
EnableAutomaticLinking	bool	false	Enable Auto Linking - This setting determines whether Cases for the given Client can be linked automatically with other Cases in the system by Link Processor. Enables automatic case linking based on configured rules.
EnableManualLinking	bool	false	Enable Manual Linking - This setting determines whether Cases for the given Client can be linked by a user manually with other Cases in the system. Allows case handlers to create case links.
ExpiredCasesAssignable	bool	false	Allow Expired Cases Assignment - This setting determines whether it is possible to assign expired cases to officers.

VRM & Vehicle Processing

Property	Type	Default	Description
PermitVrm	bool	false	Permit VRM - This setting determines whether Vehicle related data (VRM) can be added to the Client's Case. Enables vehicle registration mark processing.

Referral & Team Permissions

Property	Type	Default	Description
IsSecondReferral	bool	false	2nd Referral Work - This setting determines whether certain warrants that have already been enforced by another Enforcement Agency can be worked.
PermitWelfare / IsWelfareReferralPermitted	bool	false	Permit Welfare / Welfare Referral Permitted - This setting determines whether the Client's Cases can be worked by the Welfare team.
PermitFinancialDifficulty / IsFinancialDifficultyReferralPermitted	bool	false	Permit Financial Difficulty - This setting determines whether the Client's Cases can be worked by the Financial Difficulty Team.
ShouldOfficerRemainAssigned	bool	false	Officer Remain Assigned - This setting determines whether the assigned officer should remain on the case after completion.
ShouldOfficerRemainAssignedForEB	bool	false	Officer Remain Assigned for Exceptional Balance - This setting determines whether the officer should remain assigned specifically for Exceptional Balance cases.

Return & Reinstatement

Property	Type	Default	Description
Reinstate	bool	false	Reinstate - This setting determines whether it is possible to Reinstate a Case which is already returned. Allows returning a previously returned case back to active status.
PermitReturnCasePrematurely	bool	false	Permit Return Case Prematurely - This setting determines whether the Client's Cases can be Returned before they reach the maximum case age defined by Life

			Expectancy.
PermitReturnForPartPaidCase	bool	false	Permit Return for Part Paid Case - This setting determines whether cases with partial payments can be returned.
AutoReturnExpiredCases	bool	false	Auto Return Expired Cases - Automatically return cases when they exceed the Life Expectancy.

Goods & Assets

Property	Type	Default	Description
PermitGoodsRemoved	bool	false	Permit Goods Removed - This setting determines whether it is possible to remove Goods for the given Client's Case, represented by changing the case status to "Goods Removed" in the system.

DVLA & Address Verification

Property	Type	Default	Description
PermitDvlaEnquiries	bool	false	Permit DVLA Enquiries - This setting specifies whether DVLA (Driver and Vehicle Licensing Agency) enquiries are enabled for the given client.
IsAutomaticDvlaRecheckEnabled	bool	false	Auto Recheck After - This setting specifies if and how often automatic DVLA recheck should be performed, assuming other conditions are met.
DaysToAutomaticallyRecheckDvla	int	•	Days between automatic DVLA rechecks for ongoing address verification.
NotifyDvlaAddress	bool	false	Notify DVLA Address - Whether DVLA should be notified of address changes.

Additional Operation Settings

Property	Type	Default	Description
IncludeUnattendedReturns	bool	false	Include Unattended Returns - Include cases with unattended returns in processing.
UseLocksmithCard	bool	false	Use Locksmith Card - Enable locksmith card payment method for this client.
LetterActionWaitonHolidayDates	bool	false	Wait on Holiday Dates - Letter actions should wait for holiday dates before processing.
BureauDataType	int?	null	Bureau Data Type - [Currently Not In Use] - Type of bureau data used for this client. This field is reserved for future functionality.
CaseAgeForTbtpEnquiry	int?	null	Case Age for TBtp Enquiry - The age at which cases become eligible for TBtp (Trace Before Transfer Protocol) enquiry.

3.5.4 Finance Settings

The Finance tab controls payments, charges, fees, and payment arrangements.

Client Settings

Transport for London CC

General

Operations

Payments and Charges

Remittance and Invoicing

Client Web

Tracing

Payment Scheme

RTA Generic Payment Scheme

PDA Payment Method

Officer Payment

Payment Method

BACS

Payee

Linked Cases Money Distribution

Oldest Case First

Payment Distribution

Pro Rata

0.00

(% for Client)

Configure...

Direct Payment

☐ Apply Direct Payment as Valid Payment

Include In Direct Payment -

☐ Client Remittable Charges
☐ Allow Direct Payment to Apply on Automatically Linked Cases
☐ Marston Remittable Charges
☐ Permit Reversal of Direct Payment

Direct Payment Distribution

0.00

(% for Client)

Configure

☒ Permit Arrangement

Frequency

OneOff

Tolerance

7

☒ Generate Broken Letter

☒ Max Arrangement Length

60

days

☐ Allow custom warning on creating arrangement

☐ Interest

Charge Type

Letter Fee

Percentage

0.00

Minimum Fee

25.00

Cost Cap

9999.99

Fee Cap

9999.99

☐ Restrict payments on closed cases and inactive client

☒ Apply Fee Directly from Optimise Message

Copy Settings

Save

Close

Fee & Cost Limits

Property	Type	Default	Valid Range	Description
MinFees	decimal	Required	0-9999999999.99	Minimum Fee - The value specifies the minimum value of fee that can be applied to the client's case.
FeeCap	decimal	Required	9999999999.99	Fee Cap - This value specifies the maximum value of fee that can be applied to the client's Case.
CostCap	decimal	Required	9999999999.99	Cost Cap - This value specifies the maximum value of Cost that can be applied to the Client's Case.

Payment Configuration

Property	Type	Description
PaymentScheme	PaymentScheme	Payment Scheme - This setting determines the payment scheme for the given client.
PdaPaymentMethod	SchemeCharge	PDA Payment Method - This setting determines what sort of payment method is applied when collecting a payment and recording it via mobile endpoint.
PaymentType	int	Payment Type - The type of payment method used for paying money to the client after CPR process.
PaymentDistribution	PaymentDistribution	Payment Distribution - This setting determines the distribution of payment for the given client (e.g., Pro rata distribution).
Payee	string	Payee - Name of the payee for this client.

LinkedCasesMoneyDistribution	LinkedCasesMoneyDistribution	Linked Case Money Distribution - This setting determines how payment amounts allocated to all customer cases are distributed. Options: "Oldest case first" or "Evenly split".
PrioritisedSchemeChargeIds	List<int>	Prioritised Scheme Charges - List of scheme charge IDs in priority order for payment distribution.
Percentage	decimal	Percentage - Percentage value used for payment distribution calculations.

Payment Distribution Options (PaymentDistributionDto Enum)

Value	Enum Name	Description
0	None	No specific distribution method
1	DrakesFirst	Payment goes to Drakes first
2	ClientFirstNoCosts	Client receives payment first (excluding costs)
3	Percentage	Percentage-based distribution
4	DynamicRatio	Dynamic ratio-based distribution
5	ClientFirstCostsIncluded	Client receives payment first (including costs)
6	FeesCostsFine	Distribution order: Fees → Costs → Fine
7	ProRata	Pro-rata distribution across all charges

Linked Cases Money Distribution Options (LinkedCasesMoneyDistribution Enum)

Enum Name	Description
OldestCaseFirst	Payments are allocated to the oldest case first
EvenlySplit	Payments are split evenly across all linked cases

Arrangement Settings

Property	Type	Default	Description
PermitArrangement	bool	false	Payment Arrangement - This setting determines whether Payment Arrangements are being supported for the client's cases and what tolerances exist for each frequency.
GenerateBrokenLetter	bool	false	Generate Broken Letter - This setting determines whether a broken arrangement letter should be generated when payment arrangements fail.
MaximumArrangementLength	int?	null	Max Arrangement Length - This setting determines the maximum period for which a payment arrangement can be established (in months).
ClientFrequencyTolerances	ISet<ClientFrequencyTolerance>	•	Frequency Tolerances - Defines tolerance values (in days) for each payment frequency type. Used to determine when a payment is considered late.

Direct Payment Settings

Property	Type	Default	Description
IsDirectPaymentValid	bool	false	Direct Payment Valid - Specifies if direct payment actions are valid for this client.

IsConsiderChargeChecked	bool	false	Consider Charge Checked - Whether the "consider charge" option is checked by default.
IsDirectPaymentValidForAutomaticallyLinkedCases	bool	false	Direct Payment Valid for Auto-Linked Cases - Direct payment validation for automatically linked cases.
RestrictPaymentsOnClosedAndInactiveCases	bool	false	Restrict Payments on Closed/Inactive Cases - Whether payment restrictions apply to closed or inactive cases.

Interest Settings

Property	Type	Default	Description
Interest	bool	false	Interest Enabled - Whether interest charges are enabled for this client.
InterestPercentage	decimal?	null	Interest Percentage - The percentage rate for interest charges.
ChargeTypeId	int?	null	Charge Type ID - The type of charge applied for interest.
ApplyFeeDirectlyFromOptimiseMessage	bool	true	Apply Fee Directly from Optimise - When enabled, fees are applied directly from Optimise messaging system. (Configuration Mapping setting)
AllowPriorityChargesToBeFullyPaid	bool	false	Allow Priority Charges to be Fully Paid - When enabled, priority charges can be fully paid. (Configuration Mapping setting)

Successful Completion Fee

Property	Type	Default	Valid Range	Description
IsSuccessfulCompletionFee	bool	false	Enable Successful Completion Fee - Whether a fee is charged upon successful case completion.	
SuccessfulCompletionFee	decimal	0	max 99999.99	Successful Completion Fee Amount - Fixed fee amount charged on completion.
Percentage	decimal	0	0-100	Completion Percentage - Percentage of case value charged as fee.
CompletionFeeType	AmountValueType	•	Completion Fee Type - Whether the fee is Fixed or Percentage-based.	

3.5.5 Remittance & Invoicing Settings

The Remittance & Invoicing tab controls currency, payment runs, and invoice processes.

Client Settings

Transport for London CC

General

Operations

Payments and Charges

Remittance and Invoicing

Client Web

Tracing

Preferred Currency for Invoicing

Preferred Currency for Remittance

Client Payment Run Frequency

Daily

Officer Payment Run Frequency

Weekly

Completion Fee Invoice Frequency

Collection Commission Invoice Frequency

Client Invoice Run Frequency

Direct Payment

Include In Direct Payment -

☐ Permit Marston Remittable Charges to be Included in CIR

☐ Permit EB Negative Remittance

☐ Permit Client Invoice Run

☐ Client Paid on Case Completion

☒ Officer Paid on Case Completion

☐ Client Paid when Fine Collected

☐ Permit Hold Case to be Included In Client Payment Run and Client Invoice Run

Successful Completion Commission

☐ Applicable
☒ Numeric

0.00

☐ Percent

Exclude Payments from Calculation for

0

days

Net Client Settings

☐ Net

☐ Allow Priority Charges to be fully paid by remittable payments to Client

Rate Card

Workflow Phase Name	Rate Card
Data Cleanse	
Compliance	
Enforcement	
Case Monitoring	

Copy Settings

Save

Close

Currency Configuration

Property	Type	Description
InvoicingCurrency	CurrencyDetails	Preferred Currency for Invoicing - This setting determines the type of currency used in invoices for the given client.
RemittanceCurrency	CurrencyDetails	Preferred Currency for Remittance - This setting determines the type of currency used in remittance for the given client.

Payment Run Frequencies

Property	Type	Available Values	Description
PaymentRunFrequency	Frequency	3 weekly, 4 weekly, Daily, Fortnightly, Monthly, Quarterly, Weekly	Client Payment Run Frequency - This setting determines the frequency of the remittance process (CPR - Client Payment Remittance) for the Client's Cases.
ClientInvoiceRunFrequency	Frequency	•	Client Invoice Run Frequency - This setting determines the frequency of the invoice process (CIR - Client Invoice Run) for the Client's Cases.
CommissionInvoiceFrequency	Frequency	•	Commission Invoice Frequency - Frequency for generating commission invoices.
FeeInvoiceFrequency	Frequency	•	Fee Invoice Frequency - Frequency for generating fee invoices.

Payment Flags

Property	Type	Default	Description
ClientPaid	bool	false	Client Paid on Case Completion - This setting specifies whether the client remittance process only closes Cases (paid, returned) or also opens ones.
OfficerPaid	bool	false	Officer Paid on Case Completion - This setting specifies whether the Officer Remittance process consumes only closed cases or also open cases.
ClientPaidWhenFineCollected	bool	false	Client Paid When Fine Collected - Client receives payment when fine is collected.
IsClientInvoiceRunPermitted	bool	false	Permit Client Invoice Run - This setting specifies if Client Invoice Run process is enabled for the given client case.
IsNegativeRemittancePermitted (Permit EB negative remittance)	bool	false	Permit EB Negative Remittance - This setting specifies whether client exceptional balance can be resolved via "Reclaim with CPR" option. If true, that option is available; otherwise only "Reclaim without CPR" is available.
IsNetClient	bool	false	Net Client - Indicates if the client is a net client (payments net of fees).
IsPriorityChargesPaidAllowed	bool	false	Priority Charges Paid Allowed - Whether priority charges can be paid.
IsHoldCaseIncludedInCPRandCIR	bool	false	Hold Case Included in CPR and CIR - Whether held cases are included in payment and invoice runs.
ExcludedPaymentDays	decimal	•	Excluded Payment Days - Days excluded from payment calculations.

3.5.6 Client Web (CWA) Integration Settings

Note: Previously called "CWA Integration" in legacy systems. Now renamed to "Client Web".

Client Settings

Transport for London CC

General Operations Payments and Charges Remittance and Invoicing **Client Web** Tracing

CWA Return Codes

Selected

Client return request

Available

Advance To Legal
Agreed Monies Collected by Burlington
All Process Efforts Exhausted
Approved Premises
Arrangement agreed by client
Auto Return - Action File
Auto Return - Case Reissued

Direct Action Allowance

☒ Add Phone/Email
☒ Amend Hold
☒ Balance Adjustment
☒ Change Contact Details
☒ Client Payment

Notes and History

☒ Show History
☒ Show Notes

☐ Show Attachments In CWA In Final Status Only

Property	Description
Client Web CWA Return Codes	This setting specifies which Return Codes can be used by a client to Return Cases via Client Web Application. Defines the allowed return reasons visible to clients.
Direct Action Allowance	This setting specifies which actions performed via Client endpoint have an immediate effect on the source system. Defines which client

actions take effect immediately.

Client Web Access (CWA) Settings

Property	Type	Default	Description
ShowNotes	bool	false	Show Notes in CWA - Determines whether notes are visible in the Client Web Application.
ShowHistory	bool	false	Show History in CWA - Determines whether case history is visible in the Client Web Application.
ShowAttachmentsInCwaInFinalStatusOnly	bool	false	Show Attachments in CWA in Final Status Only - Only show attachments when case is in final status.

3.5.6 Tracing Settings

The Tracing tab controls address changes, COA (Change of Address), and trace processing.

The screenshot shows the 'Client Settings' window with the 'Tracing' tab selected. The window title is 'Client Settings' and the subtitle is 'Transport for London CC'. The tabs are: General, Operations, Payments and Charges, Remittance and Invoicing, Client Web, and Tracing. The 'Tracing' tab contains the following settings:

- Primary Address Change**
 - ☒ Set Life Exp. for CoA to 365 days
 - ☐ Allow CoA with Active Arrangement
 - Min days remaining for CoA: 42 days
 - ☐ Notify Address Change
 - Address Notification Email: [empty field]
 - ☐ Notify Officer Address Change
- CoA Settings**
 - Required action for CoA in Data Cleanse Phase: [dropdown]
 - Required action for CoA in Compliance Phase: [dropdown]
 - Required action for CoA in Enforcement Phase: [dropdown]
 - Required action for CoA in Post Enforcement Phase: [dropdown]
 - Required action for CoA in Case Monitoring Phase: [dropdown]
 - Required category for CoA in Data Cleanse Phase: COA_NOE_Marker
 - Required category for CoA in Compliance Phase: COA_NOE_Marker
 - Required category for CoA in Enforcement Phase: Allow COA To Process (Compliance Plus)
 - Required category for CoA in Post Enforcement Phase: [dropdown]
 - Required category for CoA in Case Monitoring Phase: COA_NOE_Marker
- Enforce at New Address**
 - ☒ Enforce at New Address
 - New Address Return Code: [dropdown]
 - ☒ Authorization at TEC Applicable
 - Default hold period for TEC: 999 days
 - Address Change Stage: Awaiting TEC Authorization
 - ☒ Allow fees reversal on new primary address
- Trace Cases via Workflow**
 - ☒ Trace Cases via Workflow
 - ☒ Apply COA by TBTP
 - ☐ Maximum Case age for sending Trace request: 1 days
 - ☐ Compliance Fee Auto Reverse
 - ☐ Allow workflow to block processing if linked to GANT case
 - ☒ Allow to set Previous address as Primary Address
 - ☒ Apply COA when Previous Address is set as Primary Address
 - Address Ranking Scheme: Default Ranking
 - Configure CoA Verifier Conditions: [button]

Buttons at the bottom: Copy Settings, Save, Close.

Address Change Notifications

Property	Type	Default	Description
NotifyAddressChange	bool	false	Notify Address Change - This setting determines whether, in case of successfully setting another address as primary address during the change of address, the client should be notified via email about that fact.
AddressChangeEmail	string	•	Address Change Email - Email address to receive address change notifications.

NotifyOfficerAddressChanged	bool	false	Notify Officer Address Changed - Whether the assigned officer should be notified when a case address changes.
AddressChangeStage	Stage	•	Address Change Stage - This setting determines which stages in the workflow of a case should be monitored when a successful COA is performed.

COA (Change of Address) Configuration

Property	Type	Default	Description
IsCoaWithActiveArrangementAllowed	bool	false	Allow COA with Active Arrangement - This setting determines if a Change of Address action can be performed when there is an Active Arrangement on a case.
MinimumDaysRemainingForCoa	int	•	Min Days Remaining for COA - This setting determines the minimum number of days remaining for the given client's case when a Change of Address action can be performed.
IsLifeExpectancyForCoaEnabled	bool	false	Set Life Exp. For COA to - This setting specifies to what value Case Life Expectancy should be changed when COA action triggers TEC authorisation hold for configured amount of case and the outstanding amount of days is not sufficient.
LifeExpectancyForCoa	int	•	Life Expectancy for COA - The life expectancy value to set when COA triggers the TEC authorization hold.

Enforce at New Address

Property	Type	Default	Description
EnforceAtNewAddress	bool	false	Enforce at New Address - This setting determines whether a successful COA action allows further enforcement of a case at a different primary address or triggers an immediate return.
NewAddressReturnCode	DrakeReturnCode	•	Return Code - The return code to apply when enforcing at new address is triggered.
IsTecAuthorizationApplicable	bool	false	TEC Authorization Applicable - Whether TEC (Technical Enforcement Controller) authorization applies to COA processing.
TecDefaultHoldPeriod	int	•	TEC Default Hold Period - Default hold period when TEC authorization is required.
AllowReverseFeesOnPrimaryAddressChanged	bool	false	Allow Reverse Fees on Primary Address Changed - Whether fees can be reversed when primary address changes.

COA Letter Templates

Property	Type	Description
LetterActionTemplateRequiredForCoaInDataCleanseId	int?	COA Letter Action for Data Cleanse Phase - Required action for COA in data cleanse phase.
LetterActionTemplateRequiredForCoaInComplianceId	int?	COA Letter Action for Compliance Phase - Required action for COA in compliance phase.
LetterActionTemplateRequiredForCoaInEnforcementId	int?	COA Letter Action for Enforcement Phase - Required action for COA in enforcement phase.
LetterActionTemplateRequiredForCoaInCaseMonitoringId	int?	COA Letter Action for Case Monitoring Phase - Required action for COA in case monitoring phase.
LetterRequiredActionForCoaInPostEnforcementPhaseId	int?	COA Letter Action for Post-Enforcement Phase - Required action for COA in post-enforcement phase.

CategoryRequiredForCoaInDataCleanseId	int?	COA Category for Data Cleanse - Category required for COA processing in data cleanse.
CategoryRequiredForCoaInComplianceId	int?	COA Category for Compliance - Category required for COA processing in compliance.
CategoryRequiredForCoaInEnforcementId	int?	COA Category for Enforcement - Category required for COA processing in enforcement.
CategoryRequiredForCoaInCaseMonitoringId	int?	COA Category for Case Monitoring - Category required for COA processing in case monitoring.
CategoryRequiredForCoaInPostEnforcementPhaseId	int?	COA Category for Post-Enforcement - Category required for COA in post-enforcement.

COA Verifier Conditions

The COA Verifier is a configurable set of conditions that must be checked before a Change of Address (COA) action can proceed. This is accessed via the "Configure CoA Verifier Conditions" button on the Tracing tab.

Property	Type	Description
CoAVerifierConditions	List<CoAVerifierConditionsDTO>	List of verifier conditions configured for this Client Case Type
Id	int	Unique identifier for the verifier
Name	string	Name of the verifier condition
Condition	string	The condition rule to evaluate
FailureMessage	string	Message displayed when condition fails
IsEnabled	bool	Whether this verifier condition is active
IsDeleted	bool	Whether this condition is marked as deleted
ClientCaseTypeId	int	The Client Case Type this verifier belongs to

COA Verifier Configuration (CoAExistingVerifiersDTO)

Property	Type	Default	Description
IsCaseStatusOnHoldOrGoodsRemoved	bool	false	Block COA if case status is Hold or Goods Removed
IsWelfareOrFinancialDifficulties	bool	false	Block COA if case is assigned to Welfare or Financial Difficulties team
IsAddressDirty	bool	false	Block COA if the address has been marked as dirty
IsInvalidArrestCase	bool	false	Block COA if the case is an invalid arrest case
IsClientReturnCodePresentOnPendingReturnCase	bool	false	Block COA if client return code is present on a pending return case
IsBritishForcesPostcode	bool	false	Block COA if the address has a British Forces postcode
IsCoaWithActiveArrangementAllowed	bool	false	Allow COA when active arrangement exists (override)

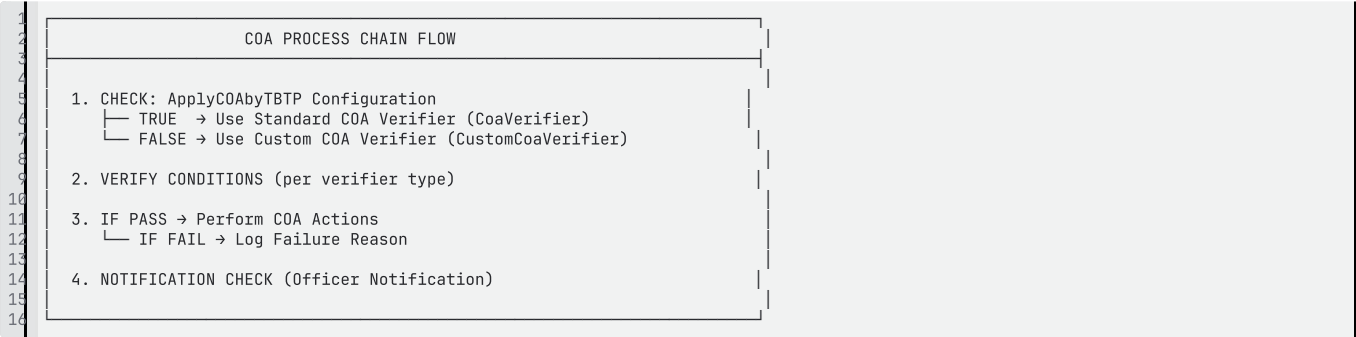
IsLifeExpectancyForCoaEnabled	bool	false	Enable life expectancy check for COA
LifeExpectancyForCoa	int	.	Life expectancy value to set for COA cases

Note: The COA Verifier conditions are stored separately from the main ClientCaseType entity and are configured via the "Configure CoA Verifier Conditions" button in the CWA application.

COA Process Chain

The COA (Change of Address) process follows a configurable chain of conditions and actions that determine whether a COA can be performed and what actions should be taken.

Process Flow Overview



COA Verification Purposes

Purpose	Description
RealChangeOfAddress	Standard COA verification for actual address changes
OfficerNotification	Verification for officer notification scenarios (allows invalid arrest cases)

Standard COA Verifier (CoaVerifier)

The standard verifier runs fixed conditions regardless of client configuration.

Conditions Checked (in order):

- 1. **Case Status Check** - Block if case status is OnHold, GoodsRemoved, or PendingReturn
- 2. **Minimum Case Life** - Block if remaining case life is insufficient for COA
- 3. **Letter Required for COA** - Block if required COA letter was not sent
- 4. **Category Required for COA** - Block if required category was not assigned
- 5. **Address Dirty Check** - Block if address is marked as dirty
- 6. **Welfare/Financial Check** - Block if case is welfare or financial difficulties
- 7. **Invalid Arrest Case** - Block if case is invalid arrest (unless OfficerNotification)
- 8. **Client Return Code** - Block if client return code is present on pending return
- 9. **Active Arrangement** - Block if COA with active arrangement is not allowed
- 10. **British Forces Postcode** - Block if British Forces postcode is not allowed

Configure CoA Verifier Conditions

Default CoA Conditions

Do not allow CoA when:

- ☒ Case Status is On-Hold or Good Removed
- ☒ Invalid Arrest case type
- ☒ Case is assigned to Welfare or Financial Difficulties Team
- ☒ Postcode is British Forces postcode
- ☒ Pending Return Case has a Client Return Code
- ☒ Address is Dirty

Do Allow CoA when:

- ☐ Allow CoA with Active Arrangement
- ☒ Set Life Exp. for CoA to days

Custom CoA Conditions

Configure Preconditions

Condition Name

Precondition Statement

Failure Reason

Validate

Save Cancel

Add

Apply Configuration Close

Custom COA Verifier (CustomCoaVerifier)

The custom verifier respects the configurable COA Verifier conditions set per Client Case Type.

Conditional Checks (respects IsEnabled flag):

Condition	When Enabled	When Disabled
Case Status OnHold/GoodsRemoved	Block if OnHold/GoodsRemoved/PendingReturn	Skip check
Address Dirty	Block if dirty	Skip check
Welfare/Financial	Block if welfare	Skip check
Invalid Arrest Case	Block if invalid arrest	Skip check
Client Return Code	Block if return code present	Skip check
Active Arrangement	Block if arrangement exists	Block always
British Forces Postcode	Block if BF postcode	Skip check

Custom Fail Reasons:

After standard conditions pass, the custom verifier also calls `VerifyCOAAndGetFailReason()` to check for additional custom failure reasons. If any custom failures exist, the result is marked as `OtherCustomFailureReasons`.

COA Fail Reasons (DefaultActionFailReason Enum)

Fail Reason	Description
None	COA verification passed
InvalidCaseStatus	Case status is OnHold, GoodsRemoved, or PendingReturn
RemainingCaseLifeIsInsufficient	Not enough days remaining on case life

AddressIsDirty	Address has been marked as dirty
InvalidCaseType	Case is an invalid arrest case
CaseIsWelfareOrFinancialDifficulties	Case assigned to Welfare/Financial Difficulties team
ClientReturnCodeNotPresentOnPendingReturn	Client return code present on pending return
CoaWithActiveArrangementNotAllowed	Active arrangement exists and COA not allowed
BritishForcesPostcodeNotAllowed	British Forces postcode not allowed
CoaLetterNotSent	Required COA letter was not sent
CoaLetterNotSentInDataCleanse	COA letter not sent in data cleanse phase
CoaLetterNotSentInCompliance	COA letter not sent in compliance phase
CoaLetterNotSentInEnforcement	COA letter not sent in enforcement phase
CoaLetterNotSentInCaseMonitoring	COA letter not sent in case monitoring phase
CoaCategoryNotAssignedToCaseInDataCleanse	Category not assigned for data cleanse
CoaCategoryNotAssignedToCaseInCompliance	Category not assigned for compliance
CoaCategoryNotAssignedToCaseInEnforcement	Category not assigned for enforcement
CoaCategoryNotAssignedToCaseInCaseMonitoring	Category not assigned for case monitoring
OtherCustomFailureReasons	Custom business rule failure
CoaWasUndoneSinceLastCoa	COA was undone since last COA was performed
LackOfAddressThatUsedToBePrimary	No address that was previously primary

COA Actions Performer

After verification passes, the following actions are performed:

Action Chain:

```
1 IF COA Conditions Satisfied:
2   └─ Reset Return Case
3   └─ Set Address As New Primary
4
5 IF No Active Arrangement:
6   └─ Log Operation Success
7
8 IF Active Arrangement:
9   └─ Log Operation Applied Without COA Actions
10  └─ Extend Case Lifetime
11
12 IF Enforce At New Address:
13   └─ Extend Case Lifetime
14   └─ Hold/Amend Hold For TEC Authorization
15
16 IF Address Change Stage Set AND Address Source != Customer:
17   └─ Hold For COA Trigger
18   └─ Reset Case
19
20 IF Fees Reversal Allowed:
21   └─ Reverse Fees (if conditions met)
22
23 Final Checks:
24   └─ Case Status After Reset Return
25   └─ Case Assignment Status
26   └─ HMCTS Case
27   └─ Address Source
28   └─ Visit History
```

Key Actions:

Action	Description
ResetReturnCase	Reset case from return status

SetAddressAsNewPrimary	Set new address as primary
ExtendCaseLifetime	Extend case life expectancy for COA
HoldOrAmendHoldForTecAuth	Hold case for TEC authorization
HoldForCoATrigger	Hold case for COA trigger
ReverseFees	Reverse fees if conditions allow
ReturnCase	Return case (if EnforceAtNewAddress not allowed)
ResetCase	Reset case status

COA Configuration Setting

Setting Name	Type	Default	Description
ApplyCOAbyTBTP	bool	false	When TRUE, use Standard COA Verifier. When FALSE, use Custom COA Verifier with client-configured conditions.

Trace Processing

Property	Type	Default	Description
TraceCasesViaWorkflow	bool	false	Trace Cases via Workflow - This setting determines whether the "Send to Trace" action in the workflow can be performed when it is present in the client's workflow.
IsPreviousAddressAllowedAsPrimary	bool	false	Previous Address Allowed as Primary - Whether a previous address can be used as the primary address.
IsTriggerCOA	bool	false	Trigger COA - Whether COA should be triggered automatically.
IsComplianceFeeAutoReversed	bool	false	Compliance Fee Auto Reversed - Whether compliance fees are automatically reversed.
IsClientInformationExchangeScheme	bool	false	Client Information Exchange Scheme - Indicates if this is a client information exchange scheme.

Expired Cases

Property	Type	Description
ExpiredCaseReturnCode	DrakeReturnCode	Return code to apply when cases expire.

3.5.8 Configuration Mappings

The Client Case Type supports dynamic configuration mappings through key-value settings.

ID	Name	Description
1	MarstonRemittableCharges	Marston Remittable Charges
2	PaymentDistributionIDDP	Payment Distribution ID for Direct Payment Scenario
3	PercentageDP	Percentage value for Payment Distribution in Percent for Direct Payment
4	SchemeChargesDP	Sceme charges selected for Direct Payment
5	ConsiderMarstonChargesCIR	Include Marston charges in CIR
6	IsReverseAllowedForDirectPayment	Direct Payment are allowed to Reverse
7	ApplyFeeDirectlyFromOptimiseMessage	Applying fee directly from Optimise Message is allowed
8	NetClient	Net Client
9	AllowPriorityChargesToBeFullyPaid	Allow Priority Charges to be fully paid by remittable payments to Client
10	AllowReturnOfCasesWithSupplementaryIndicator	Allow return of cases with supplementary indicator status
11	AllowCustomWarningOnCreatingArrangement	Allow Custom Warning On Creating Arrangement
12	AllowCustomWarningArrangementMsg	Allow Custom Warning On Creating Arrangement Message
13	ExcludedPaymentDays	No of days payment will be excluded from commission calculation
14	AddressRankingScheme	Scheme ID Selected for Address Ranking
15	LetterActionWaitonHolidayDates	Allow Letter Action Wait on HolidayDates
16	ApplyCOAbyTBTP	Apply COA by TBTP
17	ExcludeMarstonPayments	Send Only Client CPR Payment -MME
18	OfficerPaymentRunFrequency	Officer Payment Run Frequency

Configuration Mapping Entity

Property	Type	Description
ClientCaseTypeConfigurationMappingId	int	Unique identifier for the mapping
ClientCaseTypeId	int	ID of the client case type
ClientCaseTypeConfigurationId	int	ID of the configuration type
Value	string	Configuration value
IsActive	bool	Whether the configuration is active

Configuration Categories

Category	DTO Class	Description
Operation Configuration	ClientCaseTypeConfigurationMappingOperationDto	Operational settings stored as key-value pairs
Finance Configuration	ClientCaseTypeConfigurationMappingDto	Finance-related configuration
Invoicing Configuration	ClientCaseTypeConfigurationMappingInvoicingDto	Invoicing-specific settings

3.5.9 Validation Rules

The ClientCaseType entity enforces validation rules to ensure data integrity.

Required Fields

The following fields must be set:

- CaseType
- Client
- Office
- ChargingScheme
- PaymentDistribution
- Name (non-empty)
- BatchRefPrefix (non-empty)
- ContactName (non-empty)
- ContactAddress (non-empty)

Length Constraints

Field	Maximum Length
-------	----------------

Name	80 characters
BatchRefPrefix	50 characters
ContactName	256 characters
ContactAddress	500 characters
CrossReferenceEmail	512 characters
AddressChangeEmail	512 characters
ContactCentrePhoneNumber	50 characters
AutomatedLinePhoneNumber	50 characters
StandardIndustryCode	50 characters

9.3 Numeric Constraints

Field	Minimum	Maximum
MinFees	0	9999999999.99
SuccessfulCompletionFee	0	99999.99
Percentage	0	100
LifeExpectancy	1	9999
ClientLifeExpectancy	1	9999
ReturnCap	0	10000
CurrentReturnCap	0	10000
DaysToExtendCaseDueToArrangement	1	9999
DaysToExtendCaseDueToTrace	1	9999
DefaultHoldTimePeriod	1	999

Conditional Requirements

1. If **EnforceAtNewAddress** is `false`, **NewAddressReturnCode** must be `null`
2. If **ClientLifeExpectancy** is set, both **DaysToExtendCaseDueToArrangement** and **DaysToExtendCaseDueToTrace** must also be set
3. If **ReturnCap** is set, **CurrentReturnCap** must also be set within valid range
4. CrossReferenceEmail and AddressChangeEmail must be valid email formats if provided

3.5.10. Related Entities

Entity	Description	Relationship
Client	The enforcement agency or company	Many-to-One
CaseType	Case classification (HighCourt, CountyCourt, etc.)	Many-to-One
Office	Physical location/branch	Many-to-One
Brand	Brand identifier	Many-to-One
ChargingScheme	Fee calculation scheme	Many-to-One
PaymentDistribution	Payment distribution type	Many-to-One
InvoicingCurrency	Currency for invoices	Many-to-One

RemittanceCurrency	Currency for remittance	Many-to-One
Batch	Case batches	One-to-Many
ClientFrequencyTolerance	Payment frequency tolerances	One-to-Many
ClientCaseTypeConfigurationMapping	Dynamic configuration	One-to-Many
DrakeReturnCode	Status return codes	Many-to-One
SchemeCharge	Payment scheme charges	Many-to-One

Client settings for all clients have been fetched from the database and are provided in the Excel attachment below.

[ClientDetails- CMF.xlsx](#)

Tab wise details

[CCT_Settings_Tabwise_Export_20260323.xlsx](#)

Functional Requirements

FR-05.01 — API Integration

The system shall retrieve CMF parameters in real time or cached intervals.

FR-05.02 — Supported Overrides

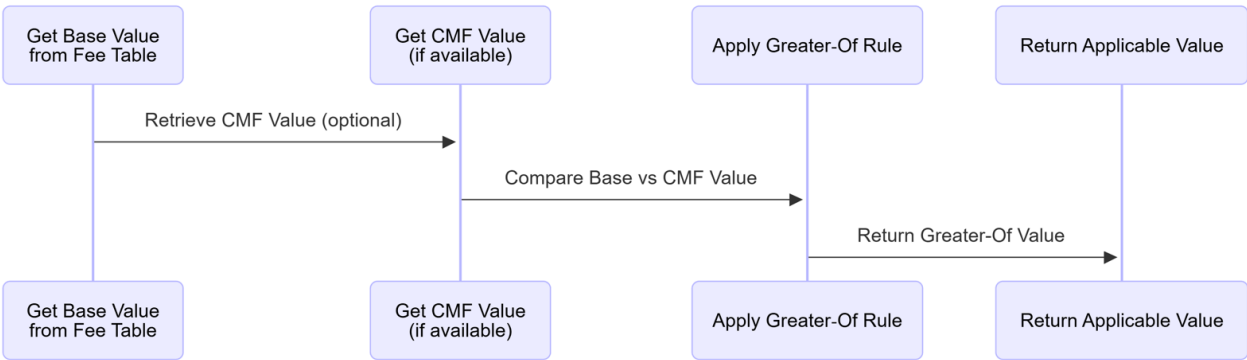
- Interest rate
- Compliance period
- Fee caps
- Custom limits for compliance extensions
- Maximum allowable override dates

FR-05.03 — Override Precedence

CMF overrides take precedence unless business rules explicitly define fallbacks.

FR-05.04 — Audit

Each override used must be recorded in the audit trail.



3.6 Keyer-Verifier Workflow (FR-06)

Description

All configuration changes must undergo a dual-control workflow: Keyer → Verifier 1 → Verifier 2.

Functional Requirements

FR-06.01 — Workflow States

- Draft

- Pending Approval (Step 1)
- Pending Approval (Step 2)
- Approved
- Rejected

FR-06.02 — Segregation of Duties

The Keyer cannot be either Verifier 1 or Verifier 2 for the same change.

FR-06.03 — Immutability

Approved changes cannot be edited or deleted.

FR-06.04 — Automatic Notifications

Email notifications must be sent at:

- Submission
- Approval 1
- Approval 2 (final)
- Rejection

FR-06.05 — Scheduled Activation

Changes become active only on their effective date.

3.7 Fee Bubble Logic (FR-07)

Description

Fee Bubbles ensure linked cases with multiple fee versions apply the correct fee scale based on the oldest active version in the bubble.

Functional Requirements

FR-07.01 — Bubble Definition

A bubble consists of cases linked by:

- Same debtor
- Same client
- Same debt group

FR-07.02 — Enforcement Fee Selection

Within a bubble:

- Existing cases use the oldest fee version
- New cases use the latest approved fee version

FR-07.03 — Version Locking

Existing cases retain their rate even if new fee versions are approved.

FR-07.04 — Audit

Audit must capture:

- Bubble ID
- Selected fee version
- Bubble composition (case list)

3.8 Optimise Parity & Conflict Handling (FR-08)

Description

The system shall maintain parity between Columbus and optimise tablets, including robust handling of offline scenarios.

Functional Requirements

FR-08.01 — Device Configuration Sync

All devices must receive updated fee configuration before applying any fee.

FR-08.02 — Offline Handling

If a device applies outdated fees offline, Columbus must:

- Identify the mismatch
- Override device values
- Flag the case for review

FR-08.03 — Direct-Apply Flag Support

Some clients allow optimise to apply fees directly; in such cases, the device **must** have the correct versioned configuration.

3.9 VAT Handling (FR-09)

FR-09.01 — Per Income Stream Rules

VAT must be applied only where applicable (e.g., High Court).

FR-09.02 — Configuration-Driven

VAT flags must be stored in configuration and applied transparently in calculations and auditing.

3.10 Versioned Activation (FR-10)

FR-10.01 — Atomic Cutover

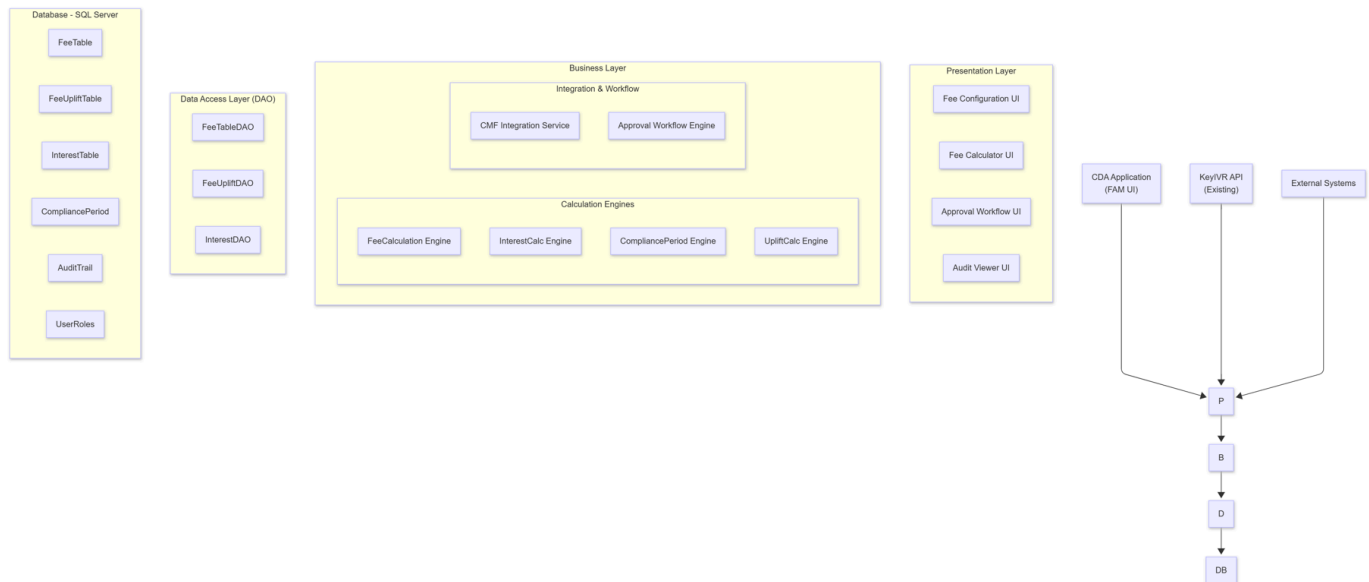
Fee, uplift, interest, and compliance configurations must activate simultaneously via configuration versioning.

FR-10.02 — No Code Deployments

All changes must be controlled through configuration and approval workflow, not software releases.

5. Architecture Design

The Fee Application Module (FAM) introduces a modern, configuration-driven architectural model that replaces legacy hardcoded logic with a governed, auditable, and highly modular structure. FAM is designed as a **centralized calculation and configuration service** that orchestrates fee determination, interest calculation, uplift rules, CMF overrides, and compliance period logic across all enforcement systems.



5.1 High-Level Architecture

FAM is composed of three primary layers:

A. Configuration Layer (SSOT)

Holds authoritative data for all fee-related rules:

- Fee Table (base fees per stage)
- Fee Uplift Table (thresholds and percentages)

- Interest Table (tiered rates with client override precedence)
- Compliance Period Table (statutory periods with CMF greater-of logic)
- CMF Integration (client-specific overrides, caps, and extension limits)
- Versioning and effective-date controls
- Dual-control Keyer/Verifier workflow metadata
- Immutable change audit records

Component	Responsibility
Fee Table	Stores base fee values for each income stream and stage; maintains multiple future-dated versions
Fee Uplift Table	Stores uplift thresholds and percentages; supports debt aggregation logic
Interest Table	Defines tiered interest rates; supports client-specific interest overrides
Compliance Period Table	Stores statutory periods; integrates with CMF for greater-of logic
CMF Integration	Dynamically retrieves client-specific overrides; supports compliance extensions
Keyer-Verifier Workflow	Governs all configuration changes; captures rationale, roles, timestamps
Audit Store	Immutable record of all configuration changes; stores digital signatures

B. Processing Layer

Executes all logic required to calculate fees deterministically:

- Fee Calculation Engine
- Uplift Evaluation Engine
- Interest Calculation Engine
- Compliance Period Engine (including greater-of rules)
- Pro-Rata Allocation Engine
- Fee Bubble Version Selector
- Rule Precedence Logic (Stage → Version → CMF)
- Optimise device conflict resolver (server-authoritative)

Component	Responsibility
Fee Calculation Engine	Runs deterministic fee selection based on income stream, stage, event date, version
Uplift Engine	Checks aggregated debt against threshold; applies uplift percentage
Interest Engine	Applies tiered interest logic; overrides with CMF where applicable
Compliance Period Engine	Determines active compliance period; triggers automatic expiry transitions
Pro-Rata Allocation Engine	Splits fees across linked cases proportionally
Fee Bubble Selector	Determines fee version across linked cases
Rule Combination Engine	Orchestrates precedence of all rules

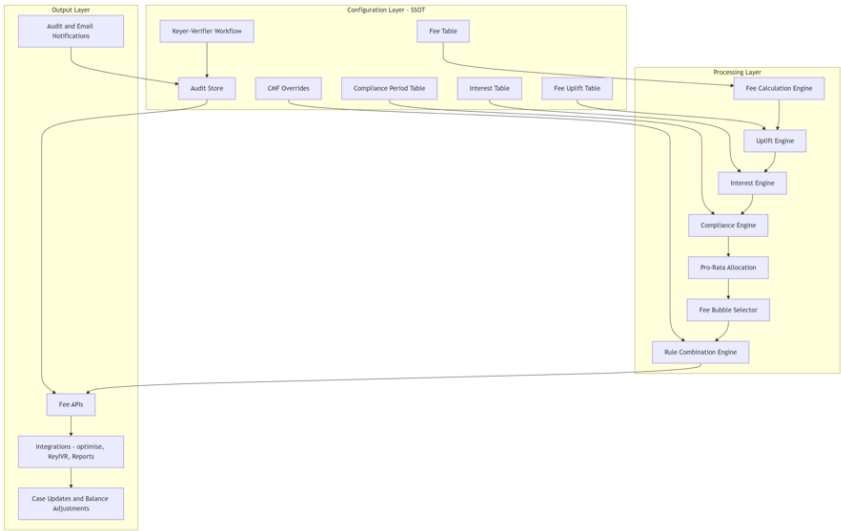
C. Output Layer

Exposes results and state transitions:

- Fee application results
- Compliance period states
- Interest breakdown
- Audit trail outputs
- Integration responses (KeyIVR, optimise, reporting, DW)
- Notifications to Audit and Governance
- Case and balance updates

Component	Responsibility
Fee APIs	Provides fee breakdown, compliance details, audit identifiers
Audit Outputs	Immutable records with digital signatures
Integration Connectors	Feeds optimise, KeyIVR, Reporting, DW
Notifications	Alerts to Audit and Governance

4.2 Logical Architecture Diagram



4.3 Component Responsibilities

4.3.1 Configuration Layer Components

Fee Table

- Stores base fee values for each income stream and stage
- Maintains multiple future-dated versions
- Used as the primary lookup for base fee determination

Fee Uplift Table

- Stores uplift thresholds and uplift percentages
- Supports debt aggregation logic
- Drives up liftable fee stages (e.g., Enforcement, S&D)

Interest Table

- Defines tiered interest rates

- Supports client-specific interest overrides

Compliance Period Table

- Stores statutory periods
- Integrates with CMF to apply greater-of logic

CMF Integration

- Dynamically retrieves client-specific overrides
- Supports compliance period extensions, caps, and fee policy exceptions

Keyer-Verifier Workflow

- Governs all configuration changes
- Captures rationale, roles, timestamps, and states

Audit Store

- Immutable record of all configuration changes
- Stores digital signatures, before/after values, and approval chains

Fee Table Management (FR-01)

The Fee Table serves as the Single Source of Truth for all enforcement fee stages.

Fee Attributes

Attribute	Type	Description
IncomeStream	String	e.g., Council Tax, RTA/TMA, High Court
FeeStage	Enum	Compliance, Enforcement, Enforcement2, Sale
FeeAmount	Decimal	Fee amount in pounds
EffectiveStartDate	DateTime	Activation start date
EffectiveEndDate	DateTime (nullable)	Activation end date (null = active)
Status	Enum	Draft, PendingApproval, Active, Retired
CreatedBy	String	User ID of creator
CreatedDate	DateTime	Creation timestamp
Verifier1	String	First approver user ID
Verifier1Date	DateTime	First approval timestamp
Verifier2	String	Second approver user ID
Verifier2Date	DateTime	Second approval timestamp
Rationale	String	Business justification
VersionTag	String	Version identifier for auditing

Supported Fee Stages

Fee Stage	Description	Applies To
Compliance	Notice of Enforcement fee	All streams
Enforcement	First enforcement attendance	All streams
Enforcement2	Second enforcement attendance	High Court only
Sale & Disposal	Goods removal/sale	All streams

Council Tax:

Fee Stage	Fee Amount	Effective Date	Effective To	Status
Compliance	£75.00	01/01/2026	31/12/2026	Active
Enforcement	£235.00	01/01/2026	31/12/2026	Active
Sale & Disposal	£425.00	01/01/2026	31/12/2026	Active

High Court Judgments:

Fee Stage	Fee Amount	Effective Date	Effective To	Status
Compliance	£75.00	01/01/2026	31/12/2026	Active
Enforcement 1	£235.00	01/01/2026	31/12/2026	Active
Enforcement 2	£455.00	01/01/2026	31/12/2026	Active
Sale & Disposal	£250.00	01/01/2026	31/12/2026	Active

4.4 Processing Layer Components

4.4.1 Fee Calculation Engine

- Runs deterministic fee selection based on:
 - Income stream
 - Fee stage
 - Case event date
 - Applicable fee version
 - CMF overrides
 - Fee bubble rules

4.4.2 Uplift Engine

- Checks if aggregated debt exceeds threshold
- Applies uplift percentage rules
- Enforces rounding behavior

Uplift Calculation Formula

1	Uplift Amount = (Aggregated Debt - Uplift Threshold) × Uplift Percentage
2	Total Fee = Base Fee + Uplift Amount

4.4.3 Interest Engine

- Applies tiered interest logic
- Overrides with CMF where applicable
- Calculates daily or monthly interest based on policy

Interest Table Structure

Attribute	Type	Description
IncomeStream	String	Target stream
BalanceThreshold	Decimal	Tier boundary
InterestRate	Decimal	Annual rate percentage
EffectiveStartDate	DateTime	Activation date
EffectiveEndDate	DateTime (nullable)	End date

The applicable tier is the **highest threshold that the debt balance meets or exceeds**.

Example (High Court Judgments):

Balance Threshold	Interest Rate	Example Debt	Applied Rate
£0.00	8.0%	£10,000	8.0%
£25,000.00	6.0%	£40,000	6.0%
£50,000.00	5.0%	£75,000	5.0%
£100,000.00	4.0%	£150,000	4.0%
£250,000.00	3.0%	£500,000	3.0%

4.4.4 Compliance Period Engine

- Determines active compliance period using greater-of logic
- Triggers automatic expiry transitions
- Applies statutory vs CMF UI states

Compliance Period Table

Attribute	Type	Description
IncomeStream	String	Target stream
BaseComplianceDays	Integer	Statutory days (e.g., 14)
EffectiveStartDate	DateTime	Activation date
EffectiveEndDate	DateTime (nullable)	End date

Greater-Of Logic

The effective compliance period is determined by:

```
1 Effective Period = MAX(Statutory Period, CMF Configured Period)
```

This ensures debtors receive the most favorable treatment.

4.4.5 Pro-Rata Allocation Engine

- Splits fees across linked cases proportionally
- Supports multi-debt and multi-stream logic
- Aligns with enforcement debt group rules

4.4.6 Fee Bubble Selector

- Determines which fee version applies across linked cases
- Ensures existing cases retain old version
- Ensures new cases use current version

4.4.7 Rule Combination Engine

- Orchestrates precedence of:
 - Fee version
 - CMF values
 - Interest overrides
 - Fee bubble outcomes
 - Uplift and base fees
 - Compliance period dictates

This ensures **same inputs → same outputs**, regardless of platform or time.

4.4.8 Fee Versioning & Effective Date Handling

Versioning Model

Each fee configuration is maintained as a versioned record with an effective start date.

Version States

State	Description
Draft	Created by Keyer, not yet submitted
Pending Approval (Step 1)	Submitted, awaiting Verifier 1
Pending Approval (Step 2)	Approved by Verifier 1, awaiting Verifier 2
Approved	Fully approved, awaiting activation
Active	Currently in effect
Retired	Superseded by newer version

Version Timeline



Supersession Logic

When a new fee version is approved:

- 1. New version is created with defined Effective Start Date
- 2. Previous active version is automatically superseded
- 3. System implicitly sets previous version's end date as day before new version's start
- 4. Only one version remains active at any given time

Example:

Version	Effective Start	Effective End
Version 1	01 Jan 2024	31 Mar 2025
Version 2	01 Apr 2025	Active (No End Date)

Future-Dated Configuration

The system supports pre-configuration of future fee versions:

- Future configurations remain inactive until effective date
- Automatic activation occurs at 00:00 on effective date
- No manual intervention required
- Ensures regulatory changes can be implemented in advance

Maximum Future Date Range

The permitted future effective date range is configurable per Debt Type:

Debt Type	Maximum Future Days
Local Authority / RTA / TMA	180 Days
High Court	365 Days
Commercial Debt	Configurable

4.5 Output Layer Components

4.5.1 Fee APIs

Provide:

- Base fee
- Uplift amount
- Interest
- VAT allocation
- Compliance period details
- Audit identifiers
- Rule justification metadata

4.5.2 Audit Outputs

- Immutable record
- Digital signature
- Value before / after
- Approval chain
- Event type

4.5.3 Integration Connectors

- optimise
- KeyIVR
- Reporting / Data Warehouse
- Client dashboards
- Internal finance systems

4.5.4 Case & Balance Updates

- Applies calculated fees
- Updates outstanding balance
- Applies expiry-driven transitions
- Flags mismatches for operational review

4.6 Architecture Principles

- **Configuration over Code:** No fee logic hardcoded.
- **Determinism:** Same inputs always produce same outputs.
- **Governance:** Dual control for all changes.
- **Auditability:** Complete, immutable audit history.
- **Traceability:** Clear rule justification for every fee.
- **Parity:** optimise and Columbus must produce identical fee outcomes.
- **Atomic Activation:** Fee, uplift, interest, and compliance changes activate together.

6. Data Models

This section defines the logical and conceptual data models that support the Fee Application Module (FAM). All core fee, uplift, interest, compliance period, override, workflow, audit, and versioning entities are represented here. These models form the foundation of the Single Source of Truth (SSOT) and ensure deterministic, auditable, and governed fee behaviour across the enforcement estate.

Each model supports:

- Effective-date versioning
- Dual-control approval workflow integration

- Immutable audit visibility
- CMF override compatibility
- Multi-stream and multi-stage extensibility

5.1 FeeTableEntity

Represents versioned base fee configurations for each fee stage and income stream.

```

1 public class FeeTableEntity
2 {
3     public int Id { get; set; }
4     public string IncomeStream { get; set; }           // e.g., Council Tax, RTA/TMA, High Court
5     public string FeeStage { get; set; }              // Compliance, Enforcement, Enforcement2, Sale
6     public decimal FeeAmount { get; set; }
7     public DateTime EffectiveStartDate { get; set; }
8     public DateTime? EffectiveEndDate { get; set; }    // null = active until superseded
9     public string Status { get; set; }                // Draft, PendingApproval, Active, Retired
10    public string CreatedBy { get; set; }
11    public DateTime CreatedDate { get; set; }
12    public string Verifier1 { get; set; }
13    public DateTime? Verifier1Date { get; set; }
14    public string Verifier2 { get; set; }
15    public DateTime? Verifier2Date { get; set; }
16    public string Rationale { get; set; }
17    public string VersionTag { get; set; }             // Used for auditing & snapshot replay
18 }

```

5.2 FeeUpliftEntity

Stores uplift thresholds, percentages, and stage associations for upliftable fee categories.

```

1 public class FeeUpliftEntity
2 {
3     public int Id { get; set; }
4     public string IncomeStream { get; set; }
5     public string FeeStage { get; set; }              // Enforcement, Enforcement2, Sale
6     public decimal UpliftThreshold { get; set; }      // Applied to aggregated debt
7     public decimal UpliftPercentage { get; set; }
8     public string RoundingMode { get; set; }          // e.g., AlwaysUp
9     public DateTime EffectiveStartDate { get; set; }
10    public DateTime? EffectiveEndDate { get; set; }
11    public string Status { get; set; }
12    public string CreatedBy { get; set; }
13    public DateTime CreatedDate { get; set; }
14    public string Verifier1 { get; set; }
15    public DateTime? Verifier1Date { get; set; }
16    public string Verifier2 { get; set; }
17    public DateTime? Verifier2Date { get; set; }
18    public string Rationale { get; set; }
19    public string VersionTag { get; set; }
20 }

```

5.3 InterestRateEntity

Defines tiered interest rules and supports CMF override precedence.

```

1 public class InterestRateEntity
2 {
3     public int Id { get; set; }
4     public string IncomeStream { get; set; }
5     public decimal BalanceThreshold { get; set; }    // Tier boundary
6     public decimal InterestRate { get; set; }        // e.g., 8.0, 6.0, 4.0
7     public DateTime EffectiveStartDate { get; set; }
8     public DateTime? EffectiveEndDate { get; set; }
9     public string Status { get; set; }
10    public string CreatedBy { get; set; }
11    public DateTime CreatedDate { get; set; }
12    public string Verifier1 { get; set; }
13    public DateTime? Verifier1Date { get; set; }
14    public string Verifier2 { get; set; }
15    public DateTime? Verifier2Date { get; set; }
16    public string Rationale { get; set; }
17    public string VersionTag { get; set; }
18 }

```

5.4 CompliancePeriodEntity

Stores statutory compliance periods per income stream and enables greater-of comparison with CMF values.

```

1 public class CompliancePeriodEntity
2 {
3     public int Id { get; set; }
4     public string IncomeStream { get; set; }
5     public int BaseComplianceDays { get; set; }      // e.g., 14 days
6     public DateTime EffectiveStartDate { get; set; }
7     public DateTime? EffectiveEndDate { get; set; }

```

```

8   public string Status { get; set; }
9   public string CreatedBy { get; set; }
10  public DateTime CreatedDate { get; set; }
11  public string Verifier1 { get; set; }
12  public DateTime? Verifier1Date { get; set; }
13  public string Verifier2 { get; set; }
14  public DateTime? Verifier2Date { get; set; }
15  public string Rationale { get; set; }
16  public string VersionTag { get; set; }
17 }

```

5.5 CMFOVERRIDEEntity (Logical Model)

CMF-driven dynamic values are not stored locally but their resolved values must be captured for audit.

```

1 public class CMFOVERRIDEEntitySnapshot
2 {
3     public int CaseId { get; set; }
4     public string ClientId { get; set; }
5     public decimal? FeeCapOverride { get; set; }
6     public int? CompliancePeriodOverride { get; set; }
7     public decimal? InterestRateOverride { get; set; }
8     public string RetrievedFromCMFAt { get; set; } // Timestamp
9 }

```

5.6 ChangeRequestEntity

Represents a fee/interest/uplift/compliance configuration change under dual-approval workflow.

```

1 public class ChangeRequestEntity
2 {
3     public int Id { get; set; }
4     public string TableName { get; set; } // FeeTable, FeeUplift, etc.
5     public int RecordId { get; set; }
6     public string ChangeType { get; set; } // Create, Update, Retire
7     public string Status { get; set; } // Draft, Pending1, Pending2, Approved, Rejected
8     public string KeyerUserId { get; set; }
9     public string Verifier1UserId { get; set; }
10    public string Verifier2UserId { get; set; }
11    public string BeforeValue { get; set; } // JSON snapshot
12    public string AfterValue { get; set; } // JSON snapshot
13    public string Rationale { get; set; }
14    public DateTime CreatedDate { get; set; }
15    public DateTime? SubmittedDate { get; set; }
16    public DateTime? Verifier1Date { get; set; }
17    public DateTime? Verifier2Date { get; set; }
18    public DateTime? RejectedDate { get; set; }
19    public string RejectionReason { get; set; }
20 }

```

5.7 AuditTrailEntity

Immutable audit records with cryptographic integrity.

```

1 public class AuditTrailEntity
2 {
3     public int Id { get; set; }
4     public string EntityType { get; set; } // FeeTable, InterestTable, CMFOVERRIDEEntity, FeeCalculation
5     public int EntityId { get; set; }
6     public string Action { get; set; } // Insert, Update, Activate, Deactivate
7     public string UserId { get; set; }
8     public string UserRole { get; set; } // Keyer, Verifier1, Verifier2, System
9     public DateTime Timestamp { get; set; }
10    public string BeforeValue { get; set; }
11    public string AfterValue { get; set; }
12    public string Rationale { get; set; }
13    public string DigitalSignature { get; set; } // Ensures non-repudiation
14    public string Hash { get; set; } // Ensures immutability
15 }

```

5.8 FeeCalculationResult (Logical Model)

Represents the output of the fee engine and is used across case updates, reporting, and audit.

```

1 public class FeeCalculationResult
2 {
3     public int CaseId { get; set; }
4     public string IncomeStream { get; set; }
5     public string FeeStage { get; set; }
6     public decimal BaseFee { get; set; }
7     public decimal? UpliftAmount { get; set; }
8     public decimal? InterestAmount { get; set; }
9     public decimal TotalFee { get; set; }
10
11    public decimal? CMFOVERRIDEEntityApplied { get; set; }
12    public string AppliedFeeVersion { get; set; }
13    public string AppliedUpliftVersion { get; set; }
14 }

```

6. API Specifications

6.1 Fee Calculation API

Request:

```
1 POST /api/FAM/CalculateFee
2 {
3   "CaseId": 12345,
4   "IncomeStream": "Council Tax",
5   "FeeStage": "Enforcement",
6   "DebtAmount": 2000.00,
7   "CalculateInterest": true,
8   "CalculationDate": "2026-02-26"
9 }
10
```

Response:

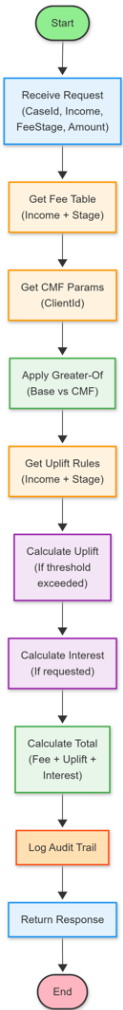
```
1 {
2   "Success": true,
3   "BaseFee": 75.00,
4   "UpliftAmount": 11.25,
5   "TotalFee": 86.25,
6   "InterestRate": 8.0,
7   "InterestAmount": 15.50,
8   "TotalAmount": 101.75,
9   "ComplianceEndDate": "2026-02-14",
10  "CalculationDetails": {
11    "UpliftThreshold": 1000.00,
12    "UpliftPercentage": 15.0,
13    "BalanceThreshold": 0.00,
14    "InterestTier": "Standard"
15  }
16 }
17
```

6.2 Fee Configuration API

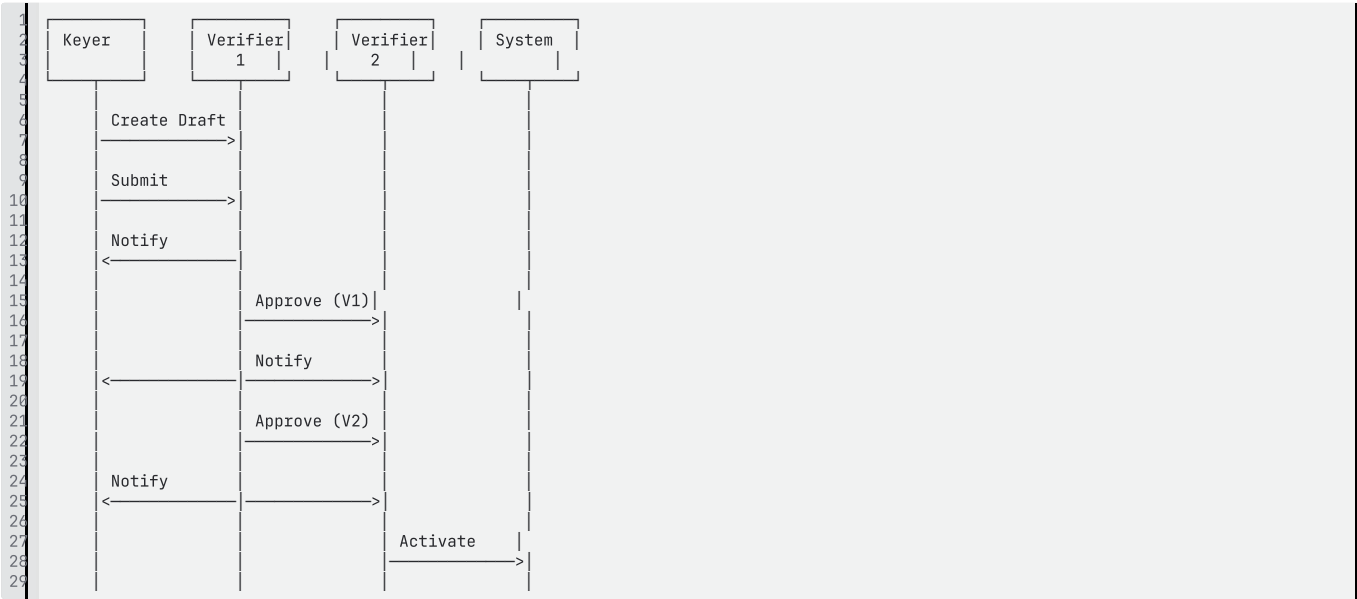
Method	Endpoint	Description
GET	/api/FAM/FeeTable/{incomeStream}	Get all fees for income stream
POST	/api/FAM/FeeTable	Create new fee (Draft)
PUT	/api/FAM/FeeTable/{id}/Submit	Submit for approval
POST	/api/FAM/FeeTable/{id}/Approve	Approve (Verifier only)
POST	/api/FAM/FeeTable/{id}/Reject	Reject (Verifier only)
GET	/api/FAM/FeeTable/History/{id}	Get approval history

7. Flow Diagrams

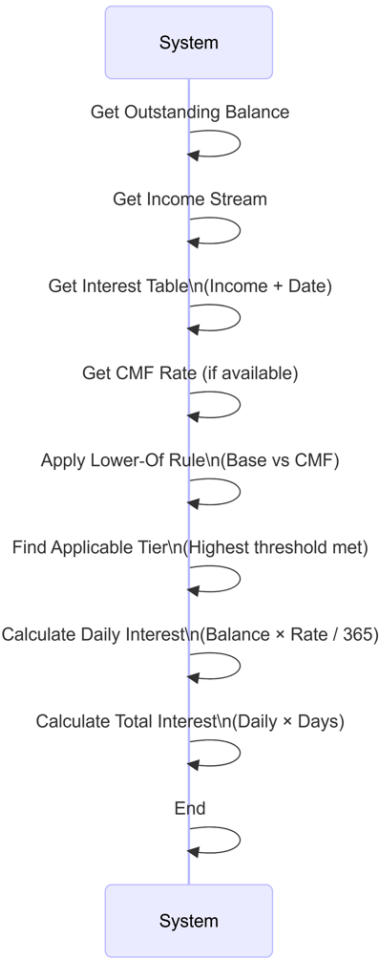
7.1 Fee Calculation Flow



7.2 Keyer-Verifier Approval Flow



7.3 Interest Calculation Flow



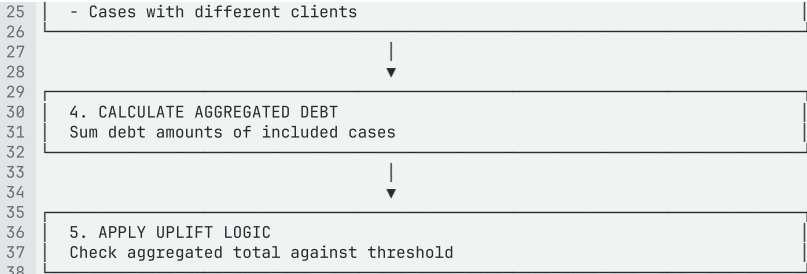
7.4 Client Override Logic

The system follows a strict precedence hierarchy for client overrides:

1	
2	
3	
4	
5	Priority 1: [HIGHEST] → Client-Specific CMF Overrides
6	Priority 2: → Interest Rate Overrides
7	Priority 3: → Compliance Period Overrides
8	Priority 4: → Fee Caps
9	Priority 5: [LOWEST] → SSOT Standard Values

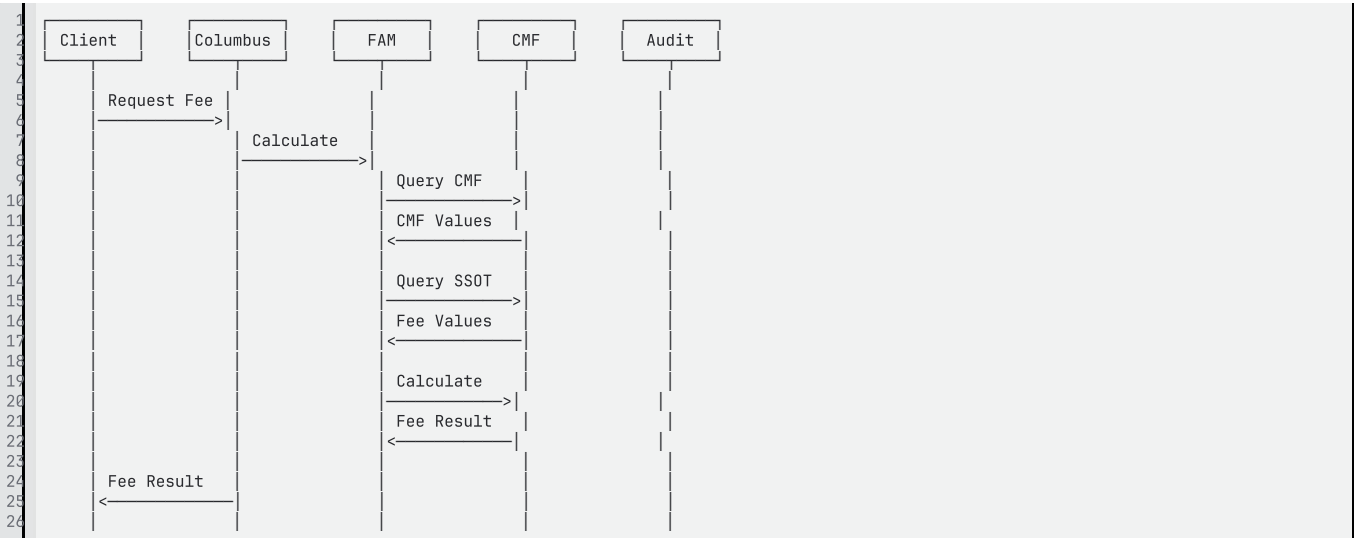
7.5 Debt Aggregation Process



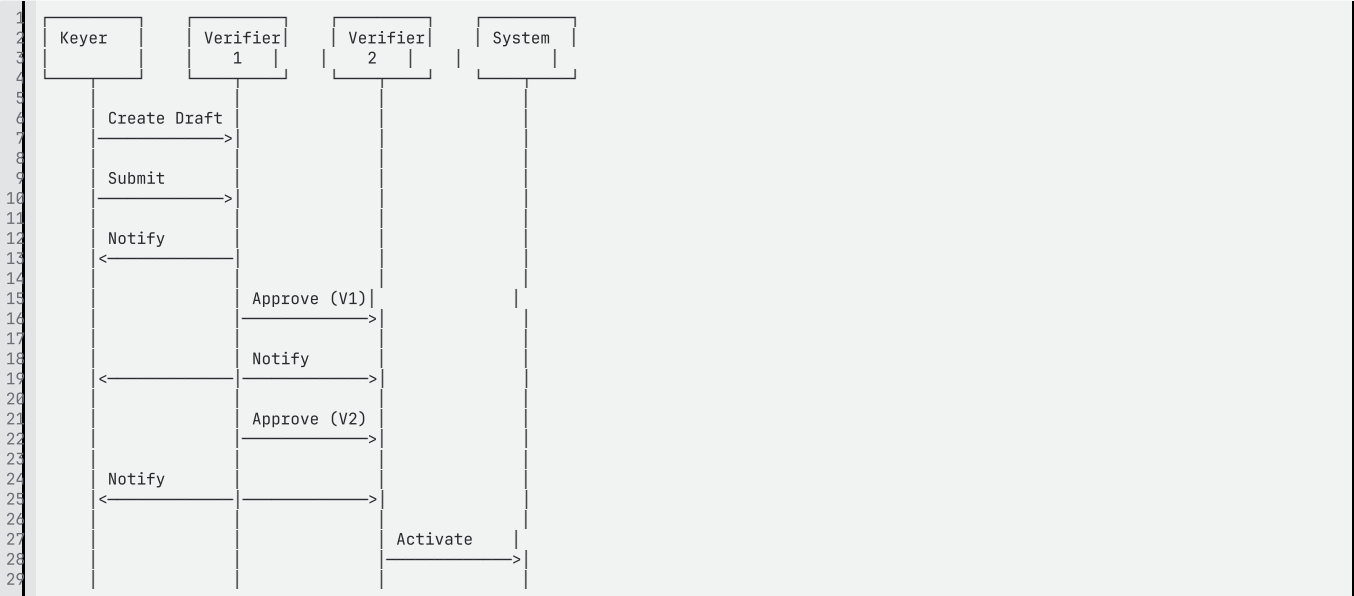


Sequence Flows for Key Operations

7.6 Fee Calculation Sequence



7.7 Configuration Approval Sequence



8. Security & Access Control

This section defines the security model, RBAC roles, permission boundaries, governance rules, and immutability constraints that apply to all Fee Application Module (FAM) operations. The objective is to ensure that all fee-related configuration, approvals, overrides, and audit records are handled securely, with segregation of duties and controlled access.

The security framework is aligned to enterprise governance standards and supports the Keyer–Verifier dual-control model required for financial/fee-related configuration.

8.1 Security Principles

FAM follows five core security principles:

1. Least Privilege Access

Users are granted only the minimum rights required to perform their responsibilities.

2. Segregation of Duties

Critical changes (fees, uplifts, interest, compliance periods) cannot be executed end-to-end by a single user.

3. Immutable Audit

All configuration actions generate immutable, cryptographically protected audit records.

4. Controlled Configuration Lifecycle

No configuration is effective until reviewed and approved through a two-stage verifier workflow.

5. Full Traceability

All changes, approvals, rejections, and rationale are visible and traceable through secure audit logs.

8.2 Roles & Responsibilities

FAM uses four RBAC roles, each with distinct capabilities to ensure governance.

1. Viewer

- Can view all configuration tables (Fee, Uplift, Interest, Compliance Period).
- Can view audit history and calculation records.
- Cannot edit, propose, or approve changes.

Intended For:

Operations, Audit, Case Review Teams.

2. Keyer (Proposer)

- Creates new configuration records (fees, uplifts, interest tiers, compliance periods).
- Edits draft versions.
- Submits changes for approval (cannot self-approve).
- Must provide rationale for all changes.
- Can view only their own pending submissions and approved configurations.

Intended For:

Fee Administrators, Policy Operations.

3. Verifier

A Verifier participates in a **two-stage approval chain**:

- **Verifier 1** (Initial Reviewer):
 - Reviews correctness, regulatory alignment, and completeness.
 - Approves or rejects the change.
- **Verifier 2** (Senior Approver):
 - Final governance check before activation.
 - Approves or rejects.

Intended For:

Governance, Assurance, Senior Operations, Finance.

4. Administrator

- Manages user accounts and RBAC assignments.
- Configures notification settings.

- Cannot create or approve fee configurations.
- Cannot bypass Keyer-Verifier workflow.

Intended For:

IT Security, Platform Administration.

8.3 Permission Matrix

Function	Viewer	Keyer	Verifier	Administrator
View Fee Table	✓	✓	✓	✓
View Fee Uplift Table	✓	✓	✓	✓
View Interest Table	✓	✓	✓	✓
View Compliance Period Table	✓	✓	✓	✓
Create Configuration Draft	✗	✓	✗	✗
Edit Draft	✗	✓	✗	✗
Submit for Approval	✗	✓	✗	✗
Approve (Verifier 1 / 2)	✗	✗	✓	✗
Reject	✗	✗	✓	✗
View Audit Records	✓	✓	✓	✓
Manage Users	✗	✗	✗	✓
Assign Roles	✗	✗	✗	✓
Configure Notifications	✗	✗	✗	✓

8.4 Segregation of Duties

To ensure governance and prevent single-actor risk:

1. The same user cannot act as Keyer and Verifier on the same change.
2. Verifier 1 and Verifier 2 must always be different individuals.
3. Administrator cannot approve fee changes.
4. All changes require rationale and dual approval before activation.
5. Effective date activation is automatic and cannot be manually overridden.

8.5 Approval Workflow Security Controls

8.5.1 State Machine Enforcement

A change request must follow strict transitions:

- **Draft → Pending Approval (Step 1)**
- **Pending Approval (Step 1) → Pending Approval (Step 2)**
- **Pending Approval (Step 2) → Approved**
- **Pending Approval (Step 1/2) → Rejected**

No state skipping is permitted.

8.5.2 Immutability

- Once approved, configuration records become **locked and cannot be edited or deleted**.
- Only new versions may be created using the formal change workflow.

8.5.3 Automatic Activation

Approved configurations become active automatically at the effective date.

This prevents:

- Premature activation
- Human error
- Manual backdating
- Silent changes or shadow deployments

8.6 Authentication & Authorization

8.6.1 Authentication

- FAM must integrate with enterprise identity providers (e.g., Azure AD).
- All access requires MFA where mandated by corporate policy.

8.6.2 Authorization

- Access must be controlled via RBAC mapped to security groups.
- Requests must be rejected if role is insufficient.
- Audit logs must capture both authenticated identity and role used.

8.7 Audit Security

FAM's audit system ensures full traceability:

- Immutable storage (append-only).
- Cryptographic hashing for tamper detection.
- Digital signatures for non-repudiation.
- Capture of:
 - User
 - Role
 - Timestamp
 - Before/after values
 - Rationale
 - Approval chain
- Audit records cannot be modified by any user, including Administrators.

8.8 Notification Security

All approval and rejection notifications must include:

- Change ID
- Table name
- Before/after value summary
- Keyer, Verifier1, Verifier2 identities
- Effective date
- Timestamp
- Non-editable message payload

Notifications are sent to:

- Keyer
- Verifiers
- Audit Function
- Governance distribution lists

8.9 Optimise & Device Security

- Device fee configurations must be validated against server-side versions.
- Offline actions require reconciliation and automatic server override.
- Outdated device configuration is treated as a security exception and flagged.
- Device actions must carry metadata for:
 - Device ID
 - User ID
 - Config version
 - Timestamps

8.10 Security Constraints

- No fee or configuration value may bypass SSOT.
- No hardcoding is allowed in any processing logic.
- No user may trigger configuration changes outside the workflow.
- No backdated effective dates are permitted outside system validation rules.
- No deletion of audit records is allowed.

9. Validation Rules

9.1 Fee Table Validation Rules

Rule ID	Rule	Description
BR-F01-001	Non-negative amounts	Fee amount must be ≥ 0
BR-F01-002	Valid effective date	Effective date must be today or future
BR-F01-003	Complete stage coverage	Each income stream must have all fee stages
BR-F01-004	No date overlaps	Duplicate effective dates not permitted
BR-F01-005	Decimal precision	Both fees and percentages to 2 decimal places
BR-F01-006	Valid status transitions	Must follow Draft → Pending → Approved → Active

9.2 Uplift Validation Rules

Rule ID	Rule	Description
BR-U01-001	Threshold ≥ 0	Uplift threshold must be non-negative
BR-U01-002	Percentage range	Uplift percentage must be 0-100%
BR-U01-003	Valid rounding	Rounding mode must be valid enum value

9.3 Interest Rate Validation Rules

Rule ID	Rule	Description
BR-I01-001	Rate range	Interest rate must be 0-100%
BR-I01-002	Ascending thresholds	Balance thresholds must be ascending
BR-I01-003	Complete coverage	No gaps between threshold ranges

9.4 Compliance Period Validation Rules

Rule ID	Rule	Description
---------	------	-------------

BR-C01-001	Positive days	Compliance period must be > 0
BR-C01-002	Maximum limit	Period cannot exceed configurable maximum

10. Business Rules

10.1 Fee Application Rules

Rule ID	Description
BR-APP-001	Fees are only applied at valid stage transitions
BR-APP-002	Compliance fee triggers on NOE generation
BR-APP-003	Enforcement fee triggers on first attendance
BR-APP-004	Sale & Disposal fee triggers on goods removal
BR-APP-005	No fees apply during compliance period (except NOE fee)

10.2 Aggregation Rules

Rule ID	Description
BR-AGG-001	Only active cases included in aggregation
BR-AGG-002	Only same debt group cases included
BR-AGG-003	Only same debtor cases included
BR-AGG-004	Closed cases excluded from aggregation

10.3 Greater-Of Rules

Rule ID	Description
BR-GT-001	Compliance period: MAX(Statutory, CMF)
BR-GT-002	Applied only during Data Cleanse and Compliance phases
BR-GT-003	Result must be captured in audit with source flag

10.4 Fee Bubble Rules

Rule ID	Description
BR-BUB-001	Bubble = Same Debtor + Same Client + Same Debt Group
BR-BUB-002	Existing cases retain oldest fee version
BR-BUB-003	New cases use current (latest) fee version
BR-BUB-004	New versions don't affect existing bubble cases

11. Audit Requirements

The Fee Application Module (FAM) must maintain a fully transparent, tamper-proof, and historically complete audit trail for all fee-related activities. This includes configuration changes, fee calculations, compliance evaluations, CMF overrides, uplift logic, interest tier application, workflow actions, optimiser overrides, and version activations.

Auditability is a critical compliance requirement, supporting operational assurance, governance oversight, regulatory scrutiny, and dispute resolution.

11.1 Audit Principles

All audit behaviour must adhere to the following principles:

1. Immutability

Audit entries must be append-only.
No record may be edited, deleted, or overwritten by any user or system process.

2. End-to-End Traceability

Every fee calculation, configuration modification, override, or workflow transition must be fully traceable to:

- The user or system component that initiated it
- The version of each configuration table applied
- The exact values used (statutory, CMF override, bubble version, uplift tier, interest tier)
- The timestamp of execution

3. Source Transparency

Audit must clearly show:

- Whether a value came from Fee Table, Uplift Table, Interest Table, CMF, Compliance Period Table
- Why that value was selected (rule justification text)

4. Non-Repudiation

All audit events must include:

- Digital signature
- Cryptographic hash
- User identity + RBAC role used

5. Automatic Capture

No manual intervention may be required for an audit event to be generated.

11.2 Audit Record Structure

Each audit entry must contain the following fields:

Field	Description
AuditId	Unique system-generated identifier
EntityType	FeeTable, FeeUplift, InterestTable, CompliancePeriod, CMFOverride, FeeCalculation, WorkflowChange
EntityId	Record being changed or referenced
Action	Insert, Update, Approve, Reject, Activate, CalculateFee, ApplyOverride
BeforeValue	JSON snapshot of previous value
AfterValue	JSON snapshot of new value
UserId	Actor generating the event
UserRole	Keyer, Verifier1, Verifier2, Viewer, System
Timestamp	UTC time of event
Rationale	Mandatory for all configuration changes
DigitalSignature	Used for non-repudiation
Hash	Ensures data integrity

VersionTag	Version snapshot used during calculation
JustificationText	Human-readable explanation of rule selection
EffectiveStartDate	(For versioned config changes)

11.3 Audit Events

FAM must generate audit entries for the following event categories:

A. Configuration Lifecycle

- Create Draft
- Edit Draft
- Submit for Approval
- Approve (Verifier 1)
- Approve (Verifier 2)
- Reject
- Automatic Effective-Date Activation

B. Fee Calculation Events

Each calculation must log:

- Base fee selected
- Uplift rule applied (yes/no), threshold, %
- Interest tier selected
- Compliance period source (Statutory / CMF / Greater-Of Result)
- Fee bubble version used
- Pro-rata allocation results
- Override corrections (optimise mismatch)
- Total fee applied
- Final justification rule

C. CMF Override Events

- Override values retrieved
- Overrides applied (e.g., interest, compliance, fee caps)
- Overrides ignored due to rule precedence

D. Workflow Actions

- Keyer submission
- Verifier 1 approval/rejection
- Verifier 2 approval/rejection
- Comments provided during rejection
- Workflow state transitions

E. Exception & Correction Events

- optimise offline mismatch correction
- Server override of incorrect device-calculated fees
- Invalid effective-date attempts
- Validation failures
- Automatic fallback logic events

11.4 Audit Notification Requirements

For governance visibility, FAM must generate **email notifications** for key events:

Notifications Triggered At

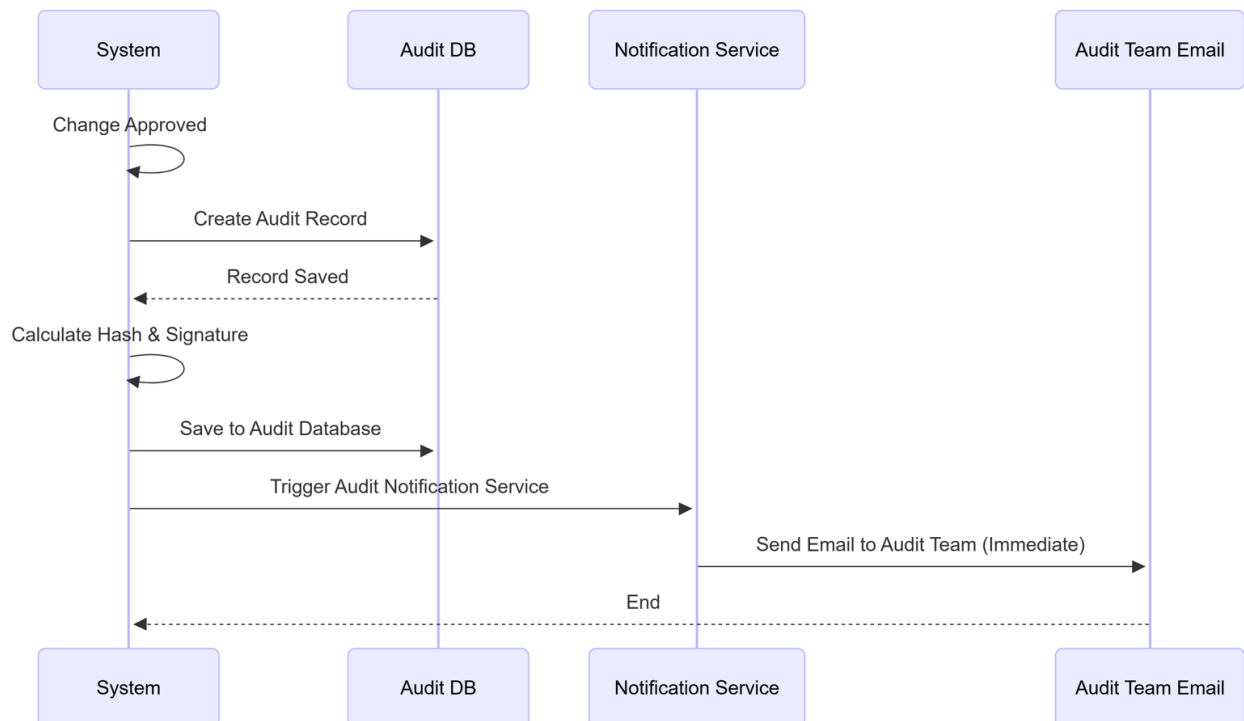
- Verifier 1 Approval
- Verifier 2 Approval (Final)
- Rejection (either verifier)
- Activation of future-dated configurations
- Critical CMF override application
- Fee bubble version conflicts
- optimise mismatch correction

Notification Recipients

- Audit Function
- Keyer
- Verifier(s)
- Governance or Policy distribution lists
- Operational leads (for exceptions)

Each notification must include:

- Change summary
- Before/after values
- Approval chain
- Timestamp
- Effective date
- Rationale
- Entity type and ID



11.5 Audit Storage Requirements

Audit data must be stored in:

- A **dedicated immutable store**

- Append-only tables
- With cryptographic protection on all rows
- Partitioned by date for performance
- Retained for **6 years from case closure**
- Backed by disaster recovery replicas

11.6 Audit for Fee Bubble Logic

Fee Bubble selection generates mandatory audit entries including:

- Bubble ID
- Case ID(s) participating
- Fee version applied
- Oldest version identified
- New case joining bubble (yes/no)
- Reason for fee version selection

11.7 Audit for Compliance Period Logic

Audit must capture:

- Statutory compliance days
- CMF compliance days
- Value chosen (greater-of result)
- Period source (Statutory/CMF)
- Start date and expiry
- Auto-transition to Enforcement

11.8 Audit for Interest Calculation

Each interest calculation must record:

- Standard tier applied
- CMF override applied (yes/no)
- Threshold crossed
- Reason interest was or wasn't applied
- Total interest added

11.9 Audit for Uplift Calculation

Audit entries must include:

- Aggregated debt value
- Cases included/excluded
- Threshold value
- Uplift percentage
- Rounding applied
- Uplift amount calculated

12. Integration Requirements

The Fee Application Module (FAM) interacts with multiple internal and external systems to ensure accurate fee calculations, real-time validation, consistent case state management, financial correctness, and operational traceability.

This section defines how FAM integrates with each dependent system and what behavior is required to maintain determinism, transparency, and parity across the enforcement ecosystem.

12.1 Integration Principles

All FAM integrations must follow these principles:

1. Server-Side Authoritative Logic

External systems (e.g., optimise, KeyIVR) may provide inputs, but FAM (Columbus) is always the single authoritative source for fee calculations.

2. Deterministic Responses

Integrations must receive the same results for the same inputs regardless of platform, time, or user.

3. Version-Aware Processing

All integrations must respect fee versioning based on effective dates.

4. SSOT Referencing Only

All returned values (fees, uplift amounts, compliance periods, interest tiers) must originate from SSOT tables, never hardcoded.

5. Audit Coverage

Every integration-driven event must generate an immutable audit trail.

6. Clear Error & Exception Handling

API consumers must receive structured errors with root-cause codes (e.g. invalid stage, invalid case, outdated device config, CMF override error).

7. Backwards Compatible API Contracts

Where integrations already rely on existing payloads, new fields must be additive, not breaking.

12.2 optimise Integration

optimise (mobile/tablet platform) is a critical integration for enforcement agents operating in the field.

12.2.1 Online Path

- optimise sends a fee request event (e.g., enforcement action).
- FAM performs full recalculation using SSOT.
- Server output overrides any device-calculated fee.
- Case is updated instantly.

12.2.2 Offline Path

If the device is offline:

- optimise applies locally cached configuration.
- On reconnection:
 - The device transmits events with its calculated amounts.
 - FAM recalculates server-side and compares results.
 - Any mismatch triggers:
 - Server override
 - Case flag for review
 - Audit entry
 - Optional operational notification

12.2.3 Configuration Sync

- Devices must sync the latest Fee, Uplift, Interest, Compliance configurations before applying fees.
- Devices unable to sync should be blocked from applying new fees.

12.2.4 Direct-Apply Flag

Some clients may require optimise to apply fees locally:

- If DirectApply = ON
→ Device *must* have the current config version
- If DirectApply = OFF
→ Only server-calculated fees are accepted

12.3 CMF Integration

The Client Maintenance File (CMF) provides client-specific overrides and parameters.

12.3.1 Required CMF Data

- Compliance period override
- Interest rate override
- Fee caps
- CMF extension limits (weekly / monthly / custom periods)
- Other client-specific fields referenced in BRD flows

12.3.2 Precedence Rule

- CMF override ALWAYS takes precedence over SSOT unless explicitly forbidden (e.g., statutory minimum).

12.3.3 Audit Traceability

All CMF-derived values must be:

- Logged in audit
- Included in API responses when relevant
- Shown in Case Details UI

12.4 KeyIVR Integration

KeyIVR relies on FAM for correct fee values during voice payments.

12.4.1 Payment Flow Integration

- KeyIVR sends current case balance and any intended payment.
- FAM returns:
 - Updated balance
 - Applicable fees
 - New fee breakdown
 - Compliance state impact

12.4.2 Arrangement Validations

If payment arrangements are created:

- FAM validates enforcement stage
- Checks compliance period state
- Validates fees applied so far

12.4.3 Real-Time Validation

KeyIVR cannot store fee values locally; all values must be retrieved real-time.

12.5 Reporting & Data Warehouse

Reporting systems must consume FAM outputs for all financial and operational insights.

12.5.1 Required Outputs

- Fee model version
- Fee bubble version
- CMF override indicators
- Compliance source (Statutory / CMF)

- Interest tier applied
- Uplift applied

12.5.2 ETL Requirements

- No historical values may be overwritten
- New Fee Model Indicator must distinguish:
 - FAM (versioned)
 - Legacy fee model (if any during transition)

12.5.3 Reconciliation Support

Audit records and FeeCalculationResults must be available for root-cause tracing.

12.6 BizTalk / Messaging Integrations

For clients who receive case updates via message queues or BizTalk:

12.6.1 Required Fields in Payloads

- Fee breakdown
- Fee version indicators
- Compliance period details
- CMF overrides
- Interest applied
- Fee bubble outcomes
- Next-stage triggers

12.6.2 Backwards Compatibility

Existing schemas must not break.

New fields must be optional or added in extended sections.

12.7 Event Grid Integration

Used for sending case and fee-related events to microservices, dashboards, or monitoring tools.

12.7.1 Event Types

- FeeApplied
- CompliancePeriodStarted
- ComplianceExpired
- FeeOverrideCorrected
- VersionActivated
- CMFOVERRIDEApplied
- MismatchDetected

12.7.2 Requirements

- Use consistent event names
- Payload must include:
 - Case ID
 - Fee breakdown
 - Applied versions
 - Timestamp
 - Rule justification

12.8 External Client Portals / CWA

External portals must display accurate fee, compliance, and balance details.

Requirements

- CWA must consume FeeCalculation API
- Must display current compliance source (Statutory/CMF)
- Must not apply any local calculation
- Must show version-aware values
- Must display breakdown of base + uplift + interest

12.9 Reconciliation & Finance Integrations

Finance tools consume fee values for CPR/OPR calculations.

Requirements

- Provide commissionable/non-commissionable fee totals
- Provide breakdowns per fee component
- Support mixed version periods
- Provide event timestamps for reconciliation windows

12.10 Error Handling & Failover Behavior

All integrations must support well-defined error structures:

Error Types

- Invalid stage
- Invalid case
- Outdated configuration
- Missing CMF data
- Effective date violation
- Version conflict
- Bubble mismatch
- Offline mismatch correction

Failover Requirements

- No fees may be applied if SSOT values are unavailable
- System must retry configuration loads automatically
- Critical failures must generate notifications to Governance and Audit

12.11 Integration Non-Functional Requirements

Performance

- Real-time fee calculations must return within acceptable latency (as per NFRs).

Scalability

- Must support estate-wide activation during cutovers.

Monitoring

- Integrations must emit telemetry for:
 - Mismatches
 - Failures
 - Exception paths
 - CMF errors
 - Device sync errors

13. Implementation Priority

This section defines the phased implementation plan for the Fee Application Module (FAM). Prioritization is based on regulatory needs, business impact, dependency flows, and delivery sequencing required to activate the full FAM capability across the enforcement estate.

Implementation follows a **phased delivery model** to minimize risk, ensure early stabilization, and allow progressive activation of interdependent components.

13.1 Phase 1 — CORE (High Priority)

Phase 1 delivers all mandatory components required for FAM to replace the legacy hard-coded fee model and introduce the new governed configuration framework.

Scope of Phase 1

- **Fee Table Management (FR-01)**

SSOT for base fees, future-dated versions, effective-date selection, and stage mapping.

- **Fee Calculation Engine (Core Logic)**

Deterministic base fee calculation with full version awareness.

- **Keyer–Verifier Workflow (FR-06)**

Dual-control approval with RBAC enforcement:

- Draft creation
- Submit
- Verifier 1 approval
- Verifier 2 approval

- **RBAC Security Model**

Roles: Viewer, Keyer, Verifier, Administrator.

- **Audit Trail – Foundation Layer**

Immutable audit entries for:

- Configuration changes
- Workflow actions
- Fee calculation snapshots

- **API Framework**

Initial set of:

- FeeCalculation API
- Fee Configuration APIs
- Audit APIs

Outcome

FAM becomes the authoritative source of base fees across all streams, with full governance, traceability, and audit foundations in place.

13.2 Phase 2 — Enhancements (Medium Priority)

Phase 2 expands FAM into a fully rule-based engine supporting uplift logic, interest tiers, compliance controls, and audit notifications.

Scope of Phase 2

- **Fee Uplift Calculation (FR-02)**

Threshold-based uplift, aggregated debt rule, rounding rules.

- **Interest Rate Configuration (FR-03)**

Tiered interest table with client override precedence.

- **Compliance Period Management (FR-04)**

Statutory periods + CMF greater-of logic

With UI differentiation and automatic expiry-to-enforcement behavior.

- **Audit Notification Engine (BO-16)**

Email notifications for approvals, rejections, overrides, and activation events.

Outcome

Fee behavior becomes fully compliant with policy, with deterministic uplift, interest logic, and intelligent compliance handling.

13.3 Phase 3 — Integration & Extensions (High Priority)

Phase 3 focuses on integrations, client alignment, and advanced rule flows required for operational readiness.

Scope of Phase 3

- **CMF Integration (FR-05)**

Overrides for:

- Interest
- Compliance periods
- Fee caps
- Client-specific limits

- **Greater-Of Logic Implementation (BO-13/14)**

Required for compliance period, interest, caps, and other override scenarios.

- **KeyIVR Integration Enhancements**

Ensuring:

- Real-time fee validation
- Correct fee breakdowns
- Accurate arrangement validations

- **Optimise Integration Stabilization**

Including:

- Device configuration sync
- Offline mismatch detection and correction
- Server override logic
- Field-safe Direct Apply behavior

Outcome

FAM becomes tightly integrated with all ecosystem platforms (CMF, optimise, KeyIVR), enabling consistent behavior across desktop, devices, payment channels, and client-facing systems.

13.4 Phase 4 — Advanced Capabilities (Optional / Priority-Based)

This phase bundles advanced rules and governance mechanisms that further strengthen system maturity.

Scope of Phase 4

- **Fee Bubble Logic (FR-07)**

Cross-case version selection with bubble ID management.

- **Pro-Rata Allocation Enhancements**

Advanced proportional fee distribution across linked cases.

- **Complex CMF Limit Configurations**

Weekly / monthly / custom-period validation flows.

- **Extended Reporting & DW Alignment**

Version indicators, fee bubble lineage, compliance-period metadata.

- **Telemetry & Monitoring Enhancements**

Device sync drift alerts
Version mismatch alerts
Performance anomaly detection

Outcome

FAM reaches full operational maturity with advanced business logic, deep traceability, and monitoring.

13.5 Implementation Principles

Across all phases:

- **Configuration before Code:**
All logic implemented in SSOT with versioning and dual control.
- **Zero-Downtime Activation:**
Version switches must occur without disrupting enforcement operations.
- **Backward Compatibility:**
Existing operational flows must not regress.
- **Full Audit Readiness:**
Every phase must support audit replay, justification logs, and forensic traceability.
- **Parallel Data Availability:**
Both legacy and FAM outputs must coexist until full migration.

14. Data Retention & Destruction

The Fee Application Module (FAM) processes and stores fee calculation data, configuration records, audit logs, CMF override details, workflow approvals, and related metadata. To support regulatory obligations, financial governance, operational forensics, and audit readiness, the system requires a robust and legally compliant data retention and destruction framework.

This section defines how long FAM-related data must be retained, how it must be protected, and how it must be securely destroyed after the retention period expires.

14.1 Retention Policy Overview

All fee-related data and audit records must be retained for:

6 years from the date of case closure

This retention period applies regardless of:

- Fee version changes
- Client-specific overrides
- Compliance extensions
- Device vs server discrepancies
- Workflow approvals and rejections

This ensures that investigations, disputes, complaints, audits, and legal reviews can be fully supported at any point during the statutory window.

14.2 Data Categories Covered

The following data categories fall under the retention policy:

1. Fee Configuration History

- Fee Table versions
- Uplift Table versions
- Interest Table versions
- Compliance period configurations
- Effective date snapshots

2. Workflow Records

- Drafts
- Approvals (Step 1 and Step 2)
- Rejections
- Rationale submitted by Keyer
- Verifier decisions

- Scheduled activation events

3. Audit Trail Entries

- Immutable audit logs
- Digital signatures
- Hash values
- Before/after values
- User role metadata
- Workflow state transitions

4. Fee Calculation Snapshots

- Base fee and uplift details
- Compliance period source selection
- Interest tier/override outcomes
- Bubble version used
- CMF overrides applied
- Pro-rata allocation outputs

5. CMF Override Retrieval Records

- Values retrieved
- Values applied/ignored
- Reason for override application

6. Integration Records

- optimise sync corrections
- Fee mismatch events
- KeyIVR fee responses
- Reporting/ETL fee outputs

14.3 Purpose of Retention

Retaining fee-related data for six years ensures:

A. Regulatory & Statutory Compliance

Supports Ministry of Justice obligations, internal governance, and legal requirements.

B. Financial Accuracy

Enables forensic investigation of:

- Incorrect fee calculations
- Disputed payments
- Historical interest and uplift logic
- Enforcement decision timelines

C. Audit Assurance

Allows auditors to replay:

- Configuration values
- Fee calculations
- Workflow decisions
- Overrides
- Compliance period selections

D. Complaints & Disputes Handling

Historical fee logic can be reconstructed, including:

- Bubble rules
- Version selection
- CMF-specific behaviors
- Uplift and interest calculations

14.4 Data Storage Requirements

Retention requires:

Secure, Immutable Storage

- Append-only architecture
- No update/delete functionality
- Cryptographically hashed records

Version Preservation

- All versions must remain fully retrievable
- Historical versions must remain intact even after new versions are activated

Indexing & Performance Considerations

- Partition audit logs by month/year
- Maintain query performance for 6 years of data

Disaster Recovery

- Backups retained and replicated
- DR environment must contain identical audit history

Access Controls

- Only authorized roles may view retained data
- No modification privileges granted to any role

14.5 Destruction Policy

After the 6-year retention period has elapsed following case closure:

Secure & Irreversible Deletion

Data must be permanently removed using:

- Cryptographically secure wipe
- No recoverability guarantees
- Destruction logs maintained

Records Eligible for Destruction

- Fee calculation snapshots
- Audit log entries
- Workflow logs
- CMF override snapshots
- Configuration state snapshots
- Fee model assignment metadata

Destruction Safeguards

Data must NOT be destroyed if:

- A legal hold exists
- An internal investigation is underway
- A complaint is active
- An auditor or regulator has requested freeze

Destruction Logging

All destruction events must be logged, including:

- Who authorized the deletion
- What data was deleted
- Timestamp
- Reason for deletion

14.6 Compliance Considerations

The retention and destruction policy ensures:

- Full alignment with financial regulations
- Low operational risk during legal reviews
- Accountability and transparency for configuration changes
- Protection of sensitive operational data
- Controlled storage growth
- Adherence to corporate data governance policies

15. Risks & Mitigations

This section identifies the key delivery, operational, compliance, and technical risks associated with the Fee Application Module (FAM), along with corresponding mitigation strategies. These risks reflect the end-to-end scope of FAM, including configuration governance, fee calculations, integrations, device parity, audit readiness, and compliance behavior.

15.1 Risk Register

Risk 1 — Activation mismatch across estate or optimise devices

Description:

Some devices or system components may not receive the latest configuration (fees, uplifts, interest tiers, compliance periods) at the cutover moment.

Impact:

Incorrect fee application, inconsistent case behavior, operational incidents.

Mitigation:

- Pre-push optimise configuration package.
- Force device sync before field operations.
- Server-side override for incorrect device-calculated fees.
- Real-time telemetry to detect drift.

Risk 2 — In-flight case ambiguity

Description:

Cases at mid-stage (e.g., in Compliance nearing expiry) may trigger inconsistent behavior if versioning/rules are not applied deterministically.

Impact:

Incorrect stage transitions, disputes, potential fee recalculation challenges.

Mitigation:

- Strict stage-based rules.
- Rule precedence engine (CMF → Statutory → Version → Bubble).
- Audit flags for compliance period source and version applied.

Risk 3 — Incorrect debt aggregation for uplift

Description:

Aggregation logic may incorrectly include/exclude linked cases or apply uplift to non-eligible debt streams.

Impact:

Misapplied uplift, financial inaccuracies.

Mitigation:

- Rule definition: same debtor + same debt group only.
- Exclude closed cases.
- Automated consistency checks in audit.

Risk 4 — Distribution & commission ripple effects**Description:**

Incorrect fees can distort payment allocation, commission outputs, or reporting.

Impact:

Financial errors for clients, officers, and reporting teams.

Mitigation:

- Validate all 7 distribution strategies using uplifted values.
- Ensure commissionable vs non-commissionable markers are accurate.
- Add Fee Model Indicator for reporting clarity.

Risk 5 — Fee Bubble version misalignment**Description:**

Linked cases may incorrectly select new versions instead of bubble-deduced oldest versions.

Impact:

Disputes, inconsistent fee application across same debtor.

Mitigation:

- Implement bubble version resolution logic.
- Log bubble composition and version selection in audit.
- Apply oldest version to all existing cases, newest for new cases.

Risk 6 — Missing or incorrect CMF overrides**Description:**

CMF may return null or outdated values, or integration issues may block overrides.

Impact:

Interest rates, compliance periods, or fee caps may be incorrectly applied.

Mitigation:

- CMF fallback logic (statutory minimums).
- Audit flags for override retrieved/ignored.
- CMF heartbeat & monitoring.

Risk 7 — Workflow bottlenecks (Keyer → Verifier)**Description:**

Approvals may be delayed or blocked if workflow queues stall.

Impact:

Critical configuration changes may not activate on time.

Mitigation:

- Escalation routes for pending approvals.
- Dashboard for outstanding change requests.
- Auto-reminders and audit notifications.

Risk 8 — Audit volume & performance

Description:

Audit-heavy components (fee calculations, overrides, config changes) may generate high-volume logs.

Impact:

Storage, indexing, and query performance degradation.

Mitigation:

- Partitioned audit tables.
- Archive strategy after threshold volumes.
- Query optimization and indexing policies.

Risk 9 — Reporting inconsistencies

Description:

If reporting models do not incorporate version indicators, CMF overrides, or uplift rules, outputs may be misaligned.

Impact:

Incorrect dashboards, financial statements, and regulatory reports.

Mitigation:

- Add Fee Model Indicator.
- Update DW ETL to incorporate version and override values.
- Validate reports during UAT.

Risk 10 — optimise offline mismatch

Description:

Devices calculating fees offline may use outdated configuration.

Impact:

Incorrect fees recorded and applied.

Mitigation:

- Server recalculation + override.
- Case flagging for operator review.
- Enforcement agent guidance during cutover windows.

Risk 11 — User error in manual transitions

Description:

Manual transitions to Enforcement or S&D could occur prematurely or incorrectly.

Impact:

Incorrect fee application, dispute risk.

Mitigation:

- UI prompts and guardrails.
- Clear messaging for compliance active periods.
- Restrictions during compliance hold windows.

Risk 12 — Pro-rata allocation misapplication

Description:

Allocation logic may miscalculate splits across multi-debt cases if thresholds or debt grouping are incorrectly applied.

Impact:

Client settlement errors, revenue distribution issues.

Mitigation:

- Rule validation: only group by same debt stream.
- Regression tests for multi-case pro-rata scenarios.
- Audit visibility of distributed per-case values.

Risk 13 — Incorrect interest tier selection

Description:

Tier selection or CMF override behavior may not correctly evaluate balance thresholds.

Impact:

Overstated/understated interest.

Mitigation:

- Tier evaluation engine with highest-threshold logic.
- CMF override precedence.
- Interest tier justification in FeeCalculationResult.

Risk 14 — Failure to enforce immutability

Description:

Configuration changes may be overwritten or altered post-approval.

Impact:

Regulatory compliance breach.

Mitigation:

- Append-only audit model.
- Locked records post-verifier approval.
- Hashing/digital signatures.

Risk 15 — Future-dated activation errors

Description:

Incorrect activation of new versions on effective date.

Impact:

Premature or delayed rollout of new fee rules.

Mitigation:

- Automatic scheduler + audit notification.
- Version simulation and UAT before activation.
- Monitoring for activation confirmation.

Risk 16 — Operational misunderstandings of new rules

Description:

Caseworkers or enforcement agents may misunderstand compliance windows, CMF rules, or uplift triggers.

Impact:

Incorrect case handling and user behavior.

Mitigation:

- Training content and quick-reference guides.
- Clear UI messaging (Statutory vs CMF mode).
- Enhanced case detail's view.

Risk 17 — Legal/Regulatory ambiguities

Description:

Incorrect interpretation of statutory rules or client agreements.

Impact:

Non-compliant fee behavior.

Mitigation:

- Legal/Policy alignment before configuration changes.
- Dual-verifier governance ensuring business correctness.
- Audit trail for traceability.

15.2 Summary of Risk Treatment Strategy

- **High-impact, High-likelihood risks** (activation, workflow, CMF, optimise) → immediate and continuous mitigation.
- **Medium risks** (user error, reporting, aggregation) → procedural guidance + test coverage.
- **Low risks** (monitoring, UI clarity, organizational alignment) → operational readiness planning.

All risks remain tracked through delivery and go-live governance boards, with clear ownership and escalation paths.

Document Control

Version	Date	Author	Changes
0.1	Mar 9, 2026	LV Team	Initial Draft
0.2	Mar 13, 2026	LV Team	Latest BRD changes (business and technical)